Advances in Emergency Demand Forecasting: A Comprehensive Systematic Review

Baiqing Sun, Ph.D.[1]; Han Zhang[2]; MinWei Deng[3]; Shuai Liu[4]; and Ruoshu Wang[5]

Abstract: Accurate emergency demand forecasting models can significantly improve decision makers' ability to make data-driven decisions during emergencies. These models help responders determine the progress of an emergency in advance and predict the potential demand for supplies, personnel, equipment, services, and other types of resources. Developing sophisticated predictive models has become vital with emerging data sources, mainly social media data and Volunteered Geographic Information (VGI). To this end, this paper first reviewed multiple reviews on emergency demand forecasting published in the same time frame and database, identifying recent advances in current forecasting models and methods and shortcomings of existing studies. At the same time, these related review studies failed to fully exploit the potential of accurate demand forecasting while failing to fully integrate cutting-edge techniques or provide a detailed categorization of demand forecasting methods. In view of these challenges, this paper focused on emergency demand forecasting and aims to fill these research gaps. By classifying, analyzing, and comparing research findings from 2020 to 2024, a comprehensive evaluation of diverse emergency demand forecasting methods is presented, drawing on publications from major databases such as Web of Science, Ei Compendex, and Google Scholar. In addition, this paper provides systematic and comprehensive emergency demand forecasts for natural disasters, general disasters, and public health emergencies, among others. It thoroughly examines critical inputs and outputs, compares various forecasting models, and integrates fundamental and technical analyses to improve the understanding and accuracy of forecasting methods. Ultimately, emergency demand forecasting is crucial in environmental research because it helps improve preparedness to respond to emergencies and can mitigate potential ecological and social impacts. **DOI: 10.1061/NHREFO.NHENG-2436.**

Author keywords: Emergency demand forecasting; Decision-making during emergencies; Machine learning; Mathematical modeling.

Introduction

As environmental issues escalate, the capacity to anticipate emergency requirements becomes progressively vital. This paper examines current research on forecasting techniques and emphasizes their utility in managing and alleviating environmental crises. In emergencies, anticipating urgent needs is essential for effective relief operations. In *Principles of Forecasting* (Armstrong 2001), a comprehensive theoretical and methodological guide to forecasting is presented, encompassing the fundamental principles from issue formulation to outcome prediction. It enumerates techniques such as role-playing and expert opinion, underscoring the significance of integrating various ways of forecasting; it also elucidates the comprehension of data processing and model implementation. This paper carefully compiles and analyzes diverse prediction methodologies, offers scientific recommendations for future study, advances foundational theories, and illustrates application outcomes through genuine case studies.

Notwithstanding the many advantages of this classic work, the specialized field of emergency management remains distinctive (Huang et al. 2021). Based on current studies in emergency demand forecasting, this direction can be roughly classified into fundamental analysis and direct technical analysis. Both classified studies aim to enhance decision-making and alleviation through future demand projections; they depend on the quantity and quality of data and employ scientific methods to analyze the data for managerial insights.

Primarily, the fundamental analysis serves as the principal research methodology; this approach evaluates prospective trends and patterns in emergency demand, predominantly influenced by economic, social, natural, environmental, political, technological, and cultural factors (Chambers and Johnson 1986; Elbanna et al. 2019; Gilissen et al. 2016; Zhang et al. 2020). Macroeconomic indicators, such as gross domestic product (GDP) and historical disaster data, can serve as significant data sources. Models designed to thoroughly and precisely predict emergency requirements incorporate multiple variables. The proliferation of variables, diverse input sources, and unstructured data present significant challenges for research. Nonetheless, the emergence of big data and machine learning technology has substantially enhanced the depth of basic analytical study (Kyrkou et al. 2022). Recent research has also addressed the complexity and obscurity of vast information volumes through the use of crawlers and natural language processing (NLP) techniques, which can efficiently extract valuable insights and elucidate the true meaning of social network platforms (Martínez-Rojas et al. 2018).

[1]Head Engineer, School of Management, Harbin Institute of Technology, 92 Xidazhi St., Nangang District, Harbin, HeilongJiang 150001, China (corresponding author). ORCID: https://orcid.org/0000-0003-3384-5029. Email: baiqingsun@hit.edu.cn

[2]Lead Researcher, School of Management, Harbin Institute of Technology, 92 Xidazhi St., Nangang District, Harbin, HeilongJiang 150001, China. Email: 20b910011@stu.hit.edu.cn

[3]Researcher, School of Management, Harbin Institute of Technology, 92 Xidazhi St., Nangang District, Harbin, HeilongJiang 150001, China. Email: 19b910026@stu.hit.edu.cn

[4]Researcher, School of Management, Harbin Institute of Technology, 92 Xidazhi St., Nangang District, Harbin, HeilongJiang 150001, China. Email: 19b310006@stu.hit.edu.cn

[5]Researcher, School of Management, Harbin Institute of Technology, 92 Xidazhi St., Nangang District, Harbin, HeilongJiang 150001, China. Email: wang_rs7@outlook.com

Note. This manuscript was published online on November 21, 2025. Discussion period open until April 21, 2026; separate discussions must be submitted for individual papers. This paper is part of the ***Natural Hazards Review***, © ASCE, ISSN 1527-6988.

Second, direct technical analysis study significantly contributes to emergency management. This research primarily relies on direct historical demand data or indicator tools to predict the extent and direction of demand fluctuations (Yan et al. 2017). A primary distinction from fundamental analysis is that technical analysis presumes all elements influencing demand are reflected in the input data; hence, direct technical analysis is advantageous for its brevity and clarity. However, it also possesses inherent limits.

Emergency demand forecasting is generally categorized into two major types. However, it lacks a precise definition and obvious boundaries. Numerous research initiatives concurrently integrate the merits of direct technical analysis with the scalability advantages of fundamental analysis studies (Toland et al. 2023); these amalgamated approaches are presently at the vanguard of research and can attain superior forecasting performance.

This review's initial section delineates the systematic review's methodology and encapsulates the findings of prior reviews. The second section delineates the pertinent research, outlining the input and output components of the predictive analysis. The third section conducts a deeper analysis of recent research, categorizing techniques such as neural networks and multiattribute decision models. In the rest of the sections, the primary obstacles of the current research and conclusions and recommendations for future endeavors are examined.

Research Methodology and Related Works

In the initial phase of the systematic review, surveys pertinent to emergency demand forecasting were identified; the Web of Science database was utilized for this investigation, focusing on articles that included "survey" as a keyword or within the abstract while excluding nonsurvey articles. Additionally, it was mandated that the abstract and keywords featured "forecasting," "prediction," "emergency," or "disaster," and only articles specifically addressing the subject of emergency management were selected. A total of 12 studies were selected, and their corresponding number of references, citation frequency, and temporal features are detailed in Table 1.

Table 1 delineates the publication date, study duration, reference count, citation frequency, pertinent issues, disaster kind, and employed technology for each research study. A thorough study reveals that these works have garnered 552 citations, all within the past 5 years. These studies concentrate on the following issues: emergency event prediction (EEP), emergency demand forecasting (EDF), impact forecasting (IFO), inventory forecasting (IF), risk assessment (RA), debris or waste forecasting (DWF), resilience (REs), emergency warning (EW), emergency materials transportation (EMT), evaluation (Ev), maritime emergency event (MEE), resource optimization (RO), monitoring (MOn), selection (SEl), needs assessment (NA), logistics assessment (LA), procurement (PRo), donation (DOn), compounding (COm), and waste prevention (WP), among others. The methodologies employed in these studies encompass, but are not restricted to, the traditional method (TM), machine learning method (ML), neural network (NN), simulation method (SM), intelligent information processing techniques (IIP), intelligent algorithm (IA), and NLP technology, among others. Emergencies can be classified into the following categories: natural disasters (ND); urban emergencies (UE); highway and logistics (HL) incidents; public health emergencies (PHE); chemical, biological, radiological, and nuclear (CBRN) catastrophes; and general disasters (GD).

These literature reviews prominently feature traditional methodologies, including experience-based analysis, Time-series analysis, fuzzy theory–based methods, Bayesian networks, multiple regression analysis, programming models, wavelet analysis, gray system theory, and various other technical tools (Huang et al. 2021; Iqbal et al. 2021; Ma et al. 2022; Tan et al. 2021; Zhu et al. 2019). The literature study typically highlights these strategies' significance due to the prevalent application of traditional methods. Huang et al. (2021) asserted that the causes, features, and types of emergencies form the foundation for categorizing prediction techniques, and classified related techniques such as traditional methods, artificial intelligence methods, and simulation methods, wherein experience-based analysis (case-based reasoning, rule-based reasoning), time-series analysis [autoregressive integrated moving average (ARIMA)], fuzzy theory–based methods (fuzzy set), and Bayesian decision theory (Bayesian network) are regarded as possessing maturity and reliability that afford them an indispensable role in emergency prediction and management. Zhu et al. (2019) confirmed the efficacy of traditional methods such as case-based reasoning (CBR), time series (ARIMA, exponential smoothing models), mathematical models (multiple linear regression, fuzzy theory–based studies), and other conventional forecasting techniques from the 1,235 articles analyzed. However, they advocate for exploring more real-time forecasting methods utilizing intelligent processing techniques, including big data mining, geographic information systems, and remote sensing (RS), to meet the demand for dynamic and purpose-driven real-time forecasting. The categorization of emergency forecasting techniques is notably subjective. Tan et al. (2021) provides a comprehensive multi-hazard overview that classifies

Table 1. List of articles

Author	Year	Start date	End date	Citations	References	Problem	Technique	Emergency type
Huang et al. (2021)	2021	2000	2001	114	181	EEP, EDF	TM, ML, NN, SM	ND, HL, UE, PHE
Zhu et al. (2019)	2019	1980	2018	82	52	EDF	TM, ML, NN, IIP, IA	ND
Tan et al. (2021)	2021	2011	2020	115	109	EDF, RA, EW, EMT, Ev	TM, ML, NN, SM, IIP, IA	ND
Heydari et al. (2022)	2022	—	—	88	109	EDF, RO	TM, SM	PHE
Ma et al. (2022)	2022	—	—	9	52	EDF	TM, NN	MEE
Chee et al. (2023)	2023	—	2021	8	137	EEP, EDF, RO, MOn	ML, NN, NLP	PHE
Hamdi et al. (2024)	2024	—	2023	8	59	EDF, IF, EMT, Mon, SEI, NA, LA, PRo, DOn, COm	—	ND, PHE, CBRN, GD
Tabata et al. (2019)	2019	—	—	43	73	DWF, WP	—	ND
Merz et al. (2020)	2020	—	—	240	410	EEP, IFO, EW	—	ND
Iqbal et al. (2021)	2021	2010	2021	30	142	All	All but NLP	All (especially ND)
Altay and Narayanan (2022)	2022	—	—	56	88	EDF, DOn, RO	TM, ML	ND, UE, GD
El-Bouri et al. (2021)	2021	—	—	36	132	EDF, RA, RO	ML, NN, NLP	PHE

several traditional methods discussed in this paper, including broad artificial intelligence (AI), fuzzy logic (FL), regression algorithms, Bayes, wavelet analysis, and expert systems as specific examples. His study also encapsulates and advocates for various techniques pertinent to the distinct phases of emergency response:

- The preparedness phase employs expert systems, artificial neural networks (ANN), support vector machines (SVM), and adaptive neuro-fuzzy inference systems (ANFIS).
- The emergency response phase utilizes robotics, ANN, FL, and SVM, among others.
- The recovery phase incorporates FL, deep learning, and multiple linear regression, among others.

Iqbal et al. (2021) identified the limitations in interdisciplinary cooperation, proposed a process-driven and demand-oriented framework, and unified technical contributions with disaster management to more precisely classify and analyze relevant constraints and address potential future contingency challenges.

Certain evaluations additionally contrast the distinctions across various contingency strategies (Chee et al. 2023; Merz et al. 2020; Zhu et al. 2019). Zhu et al. (2019) evaluate and underscore the significance of predictive methodologies such as ARIMA, CBR, and mathematical modeling while advocating for the emulation of human cognitive processes and emphasizing the need for real-time prediction strategies grounded in intelligent information processing methods. Chee et al. (2023) contended that AI methodologies surpass the efficacy of doctors and non-AI programs. Evidence maps delineate knowledge and methodological deficiencies and offer technological pathways for future research in transitioning to an emergency prehospital care scenario. In contrast to event prediction and early warning, Merz et al. (2020) underscored impact prediction, which entails projecting the intensity, location, and timing of potentially catastrophic occurrences. This methodology accurately evaluates anticipated physical damage, its distribution, human impacts, and economic losses while fostering interdisciplinary collaboration. On the other hand, certain research concentrates exclusively on a singular form of emergency scenario (Ma et al. 2022). The research question on resource allocation for maritime emergency management requires careful attention to the complexities of the maritime transportation environment. It also calls for improving emergency prediction capabilities to strengthen overall management effectiveness. Additionally, the research highlights gaps in understanding the theoretical frameworks and the mechanisms that govern the interactions between various applications, which limits the effectiveness of current approaches. Chee et al. (2023) exclusively examined the decision-making impacts of AI in pre-hospital emergency response. In contrast, Hamdi et al. (2024) thoroughly addressed the management of associated activities in disaster planning and response, proposing a framework for pharmaceutical supply management. Trash and recycling produced after earthquakes is a novel study perspective, and Tabata et al. (2019) examined pertinent policy recommendations while assessing the economic and environmental ramifications of disaster trash.

This paper concludes that numerous research initiatives are incorporated annually, as evidenced by synthesizing these 12 reviews. Merz et al. (2020) reviewed over 480 studies, Huang et al. (2021) reviewed more than 182 studies, and the other reviews reference over 50. Notwithstanding the rapid advancement of the literature, the integration of methodologies such as machine learning, neural networks, statistical modeling, image processing, artificial intelligence, and natural language processing, among others, represent prospective trends and research avenues, with scholars poised to offer further valuable contributions. This paper establishes the literature search equation and designates the Web of Science core library, EI database, and Google Scholar as the research platforms, covering the period from 2020 to 2024. This paper selects only the most pertinent 200 pieces of literature due to the extensive variety of Google Scholar search results.

This review seeks to address the subsequent research inquiries:

1. What is the efficacy of various forecasting models in emergency demand forecasting?
2. What methodologies are employed to evaluate the precision and inaccuracies of forecasting tasks?
3. What is the significance of multisource data in emergency demand forecasting?
4. What are the prevalent technologies in the realm of emergency management, and what is their importance?

This paper establishes five groups of search topic phrases: (1) emergency management, calamity, urgency; (2) forecasting model, predict, forecast; (3) resource allocation forecasting, demand estimation, demand forecasting, necessity assessment, and disaster response; (4) natural disasters, urban emergencies, highways, logistics, public health emergencies, CBRN, general disasters, man-made incidents; and (5) experience-based analysis, time series analysis, Bayesian networks, multiple regression analysis, programming models, fuzzy theory, wavelet analysis, gray system theory, machine learning methods, neural networks, simulation, intelligent information processing techniques, intelligent algorithms, natural language processing or big data, Internet of Things, smart devices, artificial intelligence. Groups 1–3 concentrate on identifying precise scientific inquiries, Group 4 ensures the selection of pertinent emergency scenarios, and Group 5 guarantees the comprehensiveness of the hunt for predictive technologies. The OR operator is utilized within the group, while the AND operator is employed between groups; the language is English, with just article and meeting selected; some disciplines (e.g., medicine, biology, and other related fields) are excluded, and preprints are removed.

Fig. 1 illustrates the selection procedure. Upon aggregating the literature from the three databases, 813 articles were identified; 53 duplicate articles were removed from the list. After reviewing the titles and abstracts, 53 documents remained, following the exclusion of 707 documents not closely related to this paper's research topic. Subsequently, six literary works were eliminated owing to irrelevance, improper inclusion, duplication, or noncompliance with requisite standards. The literature search process, which ultimately incorporated two expert-recommended works to establish a literature base comprising 49 entries.

Table 2 illustrates the distribution of journal or conference publishers. The *International Journal of Disaster Risk Reduction* appears six times, and the *International Journal of Environmental Research and Public Health* appears four times. IEEE is the preeminent conference publisher, cited six times.

Fig. 2 illustrates that the majority of studies are from 2022, with particular danger kinds, nations, time frames, and data entry modalities detailed in Table 3. Moreover, some research does not specify particular time frames or national classifications. The inputs and outputs utilized in each study are comprehensively delineated in Fig. 3.

Data Sourcing

This paper identified 17 input subtypes for the emergency demand forecasting job, organized into five groups. The subjects encompass natural and environmental sciences, social and demographic research, medical aid and incident impacts, demand management and indicators, and simulation and modeling. Fig. 4 delineates the classification framework for each data category.

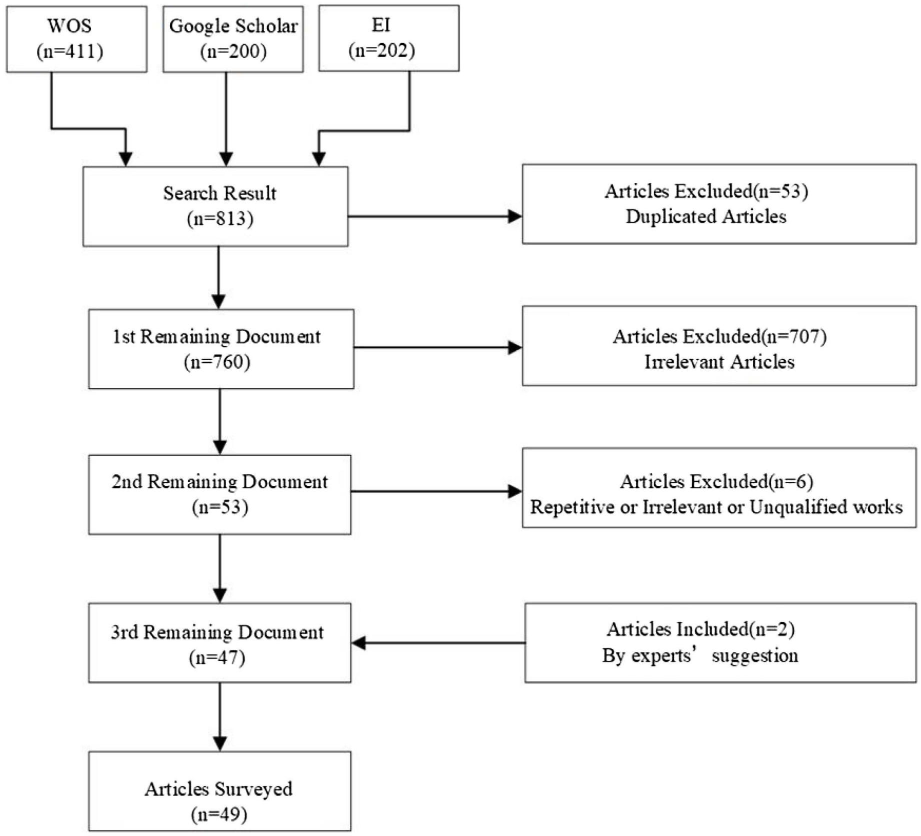

Fig. 1. Flow diagram of the selected studies.

Table 2. Grouping vocabulary

Group No.	Definition
1	Emergency management, calamity, urgency
2	Forecasting model, predict, forecast
3	Resource allocation forecasting, demand estimation, demand forecasting, necessity assessment, and disaster response
4	Natural disasters, urban emergencies, highways, logistics, public health emergencies, CBRN, general disasters, man-made incidents;
5	Experience-based analysis, time series analysis, Bayesian networks, multiple regression analysis, programming models, fuzzy theory, wavelet analysis, grey system theory, machine learning methods, neural networks, simulation, intelligent information processing techniques, intelligent algorithms, natural language processing or big data, Internet of Things, smart devices, artificial intelligence

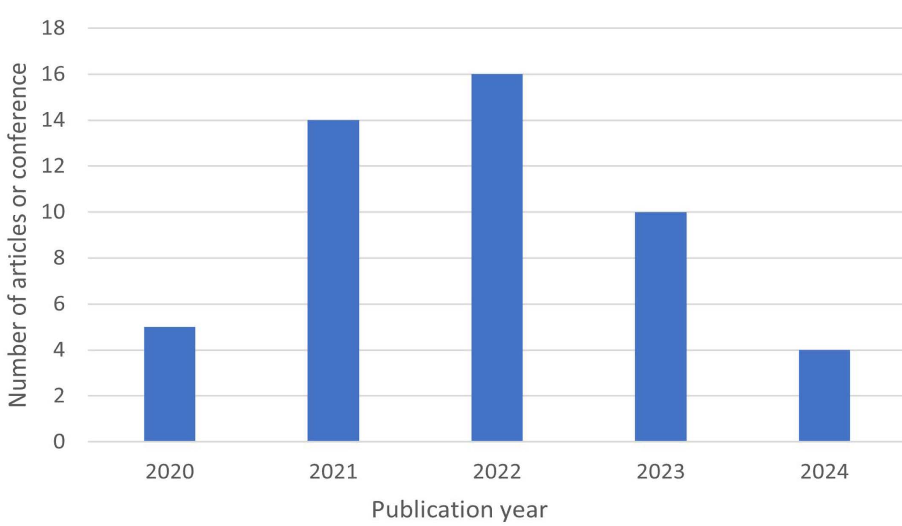

Fig. 2. Count of studies by publication year.

Table 3. Ranking of main journals and conference publishers

Journal or conference publisher	Count
International Journal of Disaster Risk Reduction	6
IEEE	6
International Journal of Environmental Research and Public Health	3
Geotechnical and Geological Engineering	2
Natural Hazards	2
IEOM Society	1
Annals of Tourism Research	1
BMC Nursing	1
Complex & Intelligent Systems	1
Complexity	1
Computer Methods and Programs in Biomedicine	1
Computers & Industrial Engineering	1
Electronics	1
Epidemiology & Infection	1
Frontiers in Sustainable Food Systems	1
Grey Systems: Theory and Application	1
IEEE Access	1
IEEE Transactions on Big Data	1
IEEE Transactions on Cybernetics	1
IEEE Transactions on Emerging Topics in Computing	1
International Journal of Geographical Information Science	1
International Journal of Innovative Computing, Information and Control	1
International Journal of Production Economics	1
International Journal of Production Research	1
IOP Publishing	1
ISA Transactions	1
Journal of Humanitarian Logistics and Supply Chain Management	1
Journal of Industrial Information Integration	1
Lancet Global Health	1
Neural Computing & Applications	1
Operations Research Perspectives	1
Plos One	1
Socioeconomic Planning Sciences	1
The American Journal of Emergency Medicine	1
Value In Health	1
Total	49

Natural and Environmental Sciences

This category encompasses three types of input data: geology, meteorology, and hydrology. These are fundamental prerequisites for precise emergency demand forecasting and augmenting its efficacy. The input data of the geology type are utilized to comprehend and examine the geological environment, encompassing focal depth, topographical characteristics, geographical context, magnitude, and other pertinent elements (Li et al. 2022b; Hao et al. 2023; Oktarina et al. 2020; Hu et al. 2021a). According to the reviewed literature, such studies were identified to systematically arrange and evaluate historical events or integrate numerous data sources to forecast the occurrence and implications of future calamities collectively. Oktarina et al. (2020) analyzed 123 comprehensive earthquake datasets from Indonesia's DIBI database and accurately forecasted fatalities, injuries, property damage, and emergency response requirements utilizing artificial neural network methodologies. Liang et al. (2024) introduced a method for integrating information flow via a shared platform, amalgamating disaster-related data with multiscale demographic information to ascertain the distribution of affected populations and shelters while systematically analyzing the allocation of emergency supplies. This approach seeks to optimize the method's efficacy by maximizing and integrating information resources, including information flow and logistics. To examine impact variables more thoroughly and save response time, Hu et al. (2021a) innovatively combined meteorological and geographic information system (GIS) data to evaluate the effects of disasters on grid equipment and identify pathways for postdisaster material demand and distribution. Nevertheless, the studies within this category differ according to their research objectives. Landslides and mudslides (La et al. 2022b) address one dimension, whereas fault lines and seismic intensity (Fang et al. 2021) investigate a different aspect. Intensity is utilized to forecast the distribution of rescue personnel and medical supplies.

Meteorological data are prevalent in the literature and typically encompass variables such as wind force, speed, precipitation, intensity, and temperature (Song et al. 2023; Ramgopal et al. 2021; Hu et al. 2021a; Li et al. 2022b; Li and Liu 2023). Analysis of the literature presented in Table 3 reveals that numerous studies employ

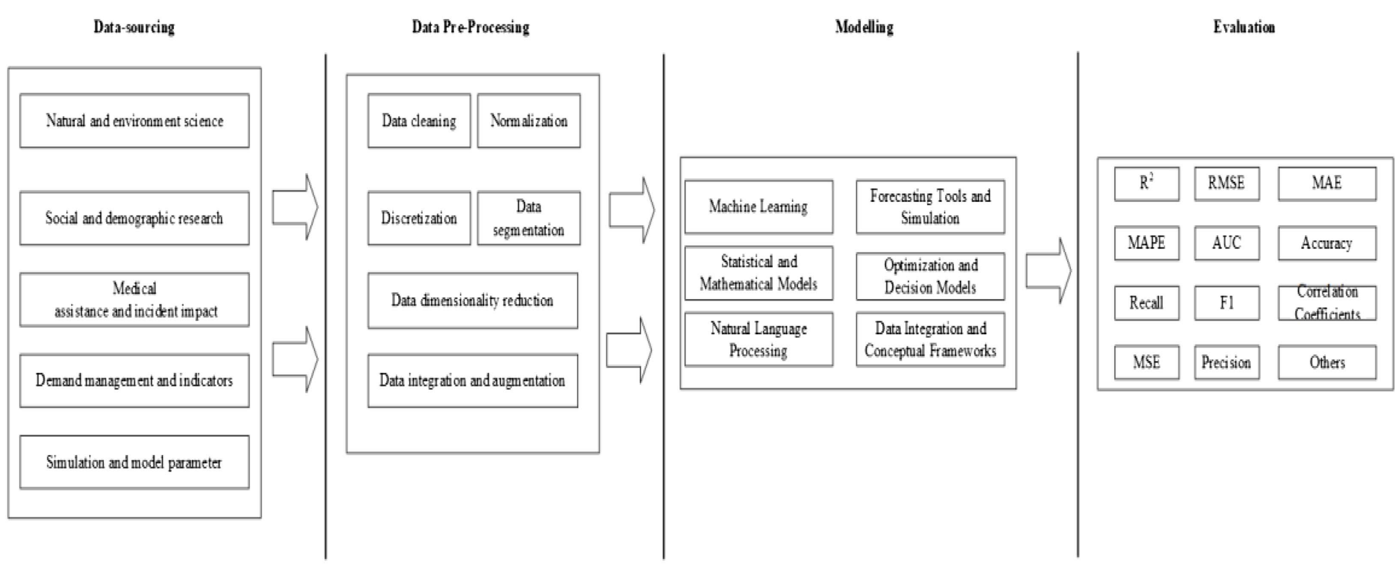

Fig. 3. Phases of the emergency demand forecasting.

Nat. Hazards Rev., 2026, 27(1): 03125003

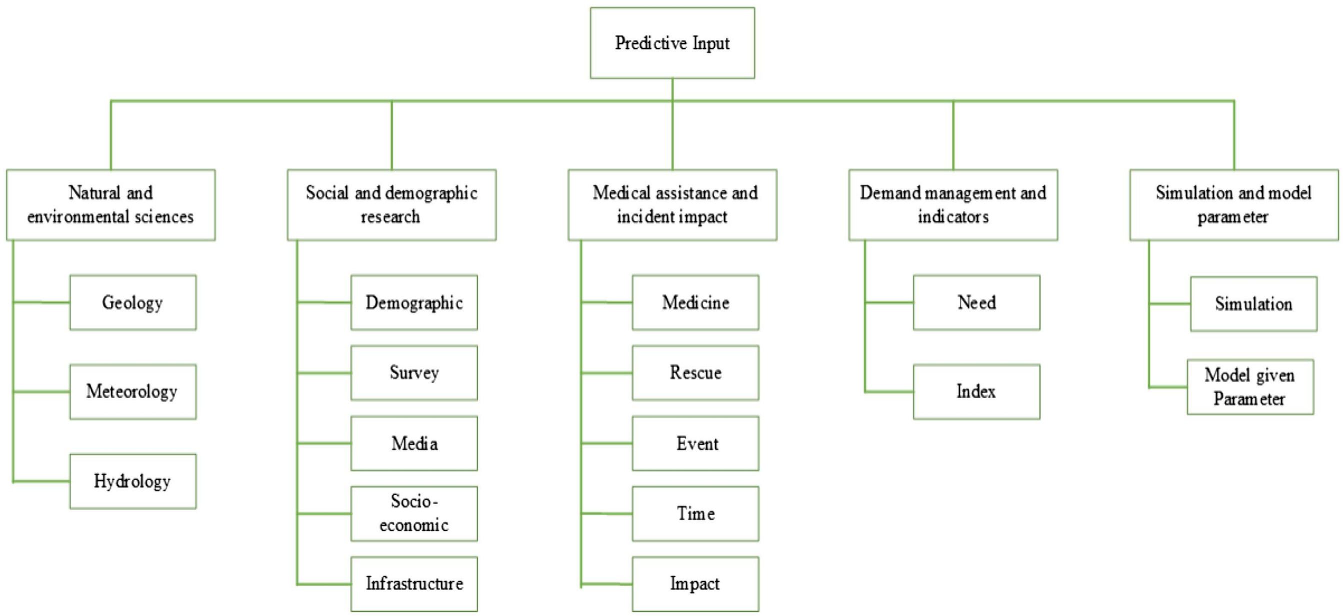

Fig. 4. Classifications of input data type.

meteorological data as an input variable for emergency demand forecasting, favoring integrating multisourced data and applying machine learning and deep learning models to predict emergency demand comprehensively. Nonetheless, there remain minor distinctions among these articles. Li et al. (2022b) concentrated on the quantity of rescue personnel in healthcare contexts, while Zheng et al. (2022) emphasized demand forecasting in natural disaster situations. Hu et al. (2021a) concentrated on the cumulative impacts of meteorological data in temporal analysis, while Fei and Wang (2022) used a cohesive method for allocating meteorological data to a singular event.

In contrast to the preceding two categories, just one study clearly referenced the hydrology kind of input data, which incorporated rainfall data, flood magnitude, and predictive levels. It utilized input variables for forecasting resource demand during floods and employed a combination of analytic hierarchy process (AHP) and system dynamics methodologies to extract the data's potential value.

Social and Demographic Research

This category primarily focuses on social and demographic attributes, with common input data including population density, income, educational attainment, crime statistics, and population migration (Li et al. 2022a; Song et al. 2023; Jiang et al. 2021, 2024; Jin et al. 2021). Initially, studies utilizing demographic data as input variables for emergency needs typically source relevant information from official statistics, corporate platforms, or surveys. Lin et al. (2020) sought to acquire population density data and dynamically forecast the need for urban flood relief materials via Baidu's big data platform. Furthermore, Chen et al. (2022) proposed that demographic fluctuations can be effectively monitored by integrating diverse data sources, including social media and survey data. Marzook et al. (2023) underscored the creation of AI-driven mobile applications informed by variables like patient density, which can be inferred through real-time intensive care unit (ICU) monitoring.

Conversely, typical input data associated with media and surveys include tweets, news articles, voluntary geographic information, questionnaires from medical personnel, and disaster experiences (Jiang et al. 2021; Marzook et al. 2023; Nguyen et al. 2022, 2023; Zhang et al. 2021; Chen et al. 2022; Liang et al. 2024; Gao et al. 2022; Chen and Ji 2022), which can be obtained with significant timeliness or comprehensiveness. They serve as a crucial information access channel that significantly enhances public engagement and the efficacy of emergency response. Certain researchers prefer telephone interviews for collecting survey data, but others may advocate for questionnaire surveys (Jiang et al. 2021). The study prioritized the immediacy and accessibility of data while analyzing the target audience for data collection. It concentrated on assessing public demand for certain catastrophic occurrences in comparison to the studies conducted by Ramgopal et al. (2021). Upon acquiring this data type, Nguyen et al. (2023) employed sentiment analysis, whereas Chen and Ji (2021) utilized keyword matching to process and analyze it as a foundation for subsequent modeling.

Given the many interconnections between society and the economy, a unilateral analysis of the input data from a single aspect seems inadequate. Consequently, the socioeconomic category typically encompasses components such as gross national product (GNP), land utilization, economic detriment, Consumer Price Index (CPI), fiscal budget, and community vulnerability assessment (Zheng et al. 2022; Jin et al. 2021; Liang et al. 2024; Zhu et al. 2022; Toland et al. 2023; Biswas et al. 2024; Fei and Wang 2022; Jiang et al. 2024). These features signify distinct concepts, such as community vulnerability and social vulnerability interaction, which facilitate the evaluation of catastrophe victims' conditions and rehabilitation processes (Edejer et al. 2020).

Simultaneously, market resources and donor funding dictate resource allocation (Jiang et al. 2024). Furthermore, urbanization correlates with the concentration of population and infrastructure inside urban areas (Zheng et al. 2022), while land use is influenced by the interplay between population distribution and resource availability (Jin et al. 2021). In conclusion, infrastructure development is fundamental for the proper functioning of society; robust infrastructure can endure calamities, bolster the resilience of social emergency responses, and ensure livelihood security for disaster victims. The resilience of power, water supply, and traffic light systems is a primary subject of emergency management research, warranting the utilization of pertinent infrastructure conditions as inputs. Typical infrastructure data input types encompass road network density, failure rate, transportation system damage, and fire department location (Song et al. 2023; Hu et al. 2021a; Toland et al. 2023; Mallouhy et al. 2022). The research gathers the locations of vital infrastructure (Ma et al. 2023). Furthermore, real-time data must be acquired using sensors and social media to formulate emergency response methods.

Medical Assistance and Incident Impact

Medicine is a key input element in emergency demand forecasting, typically associated with medical services (Jiang et al. 2024; Hezam 2022; Caro et al. 2021; Cheng et al. 2021). Li et al. (2022b) and Marzook et al. (2023) concurrently underscored the significance of ICU beds and employ various methodologies to track fluctuations in healthcare resources and demand. Kamar et al. (2021) and Hezam (2022) forecasted the impact of infectious disease transmission to ascertain actual demand; additionally, Wang and Jia (2023) took into account the existence of asymptomatic sick patients based on prior research. The study by Fang et al. (2021) was particularly distinctive because it examined alterations in injuries and event conditions and formulated a spectrum of diseases to manage health issues in catastrophe contexts comprehensively.

This paper additionally classifies another category of input data as rescue, emphasizing its significance in predictive analysis; Wang and Jia (2023) employed intuitionalist fuzzy set summation and gray-Markov techniques, utilizing infected population data to forecast the demand for emergency supplies by safety stock theory. The investigations by La et al. (2022a, b) underscored the study of a scenario-task-based method, with the former focusing on demand forecasting and implementing various rescue themes. Simultaneously, the latter emphasizes the predictive analysis of life-saving equipment for rescue operations. While there are similarities between the two, there are minor distinctions in the emphasis of the study and particular approaches.

This paper categorizes closely related data into event, time, and impact classifications. The event denotes the time, location, and causation of historical occurrences; specifically, this type of article underscores the significance of emergencies, with 29 highlighting the event's causation (Mallouhy et al. 2022; Biswas et al. 2024). Time refers to the temporal aspects related to forecasting, such as real time and duration of wait, with reply time explicitly denoting the period of waiting and reaction (Ramgopal et al. 2021). Impact pertains to the effects on society at large, including fatalities, economic losses, and similar factors (Li et al. 2022a). Fatalities and casualties serve as critical intermediate variables for assessing emergency needs. Economic loss is also referenced in Nguyen et al. (2022) and Chen et al. (2022).

Demand Management and Indicators

In emergency management, demand forecasting is essential for effective relief efforts. This paper classifies demand-related data as need type, encompassing material classifications, quantities, and demand volatility data (Fang et al. 2021; Nguyen et al. 2023). Initially, the disorders were examined in the research, followed by identifying appropriate medical demands depending on the disease kind. Zhu et al. (2022) did a direct qualitative and quantitative analysis of 14 essential food items during the earthquake. These are all data kinds directly associated with requirements.

The index type of data is comprehensively documented in Table 4. This table indicates that the indices primarily pertain to equipment and infrastructure, geology, societal vulnerability, medical requirements, congestion, and severity. Hu et al. (2021a) employed the equipment and terrain indexes to assess equipment vulnerability and terrain complexity. Hezam (2022) referenced the Oxford stringency index and adjusted it to evaluate the extent of mitigation measures for the COVID-19 pandemic. In a public health crisis, Cheng et al. (2021) employed emergency department crowding scores to assess and measure crowding. Ramgopal et al. (2021) employed the degree of intolerance to pain to assess the pain tolerance of the injured, facilitating prompt and dynamic demand prediction.

Simulation and Model Parameter

Simulation data generally include desktop walkthrough data, time-sharing simulations, population simulation data, and resident demographic characteristics (Li et al. 2022a; Jiang et al. 2021; Geng et al. 2024). Simulation data can represent various emergencies and provide real-time modifications and dynamic adaptations to swiftly evolving situations. Li et al. (2022a) employed historical instances, real-time disaster data, time-sharing simulation data, temporal data, and rough set theory to estimate attributes for forecasting the number of fatalities using a SVM. Ultimately, a segment of input class data associated with the model-specified parameter category is mainly utilized for algorithm calibration and modeling enhancement, aiding in resolving intricate decision-making challenges.

Data Preprocessing

Before inputting diverse data sources into a predictive model, numerous preprocessing techniques are typically utilized to guarantee the precision and reliability of the projected outcomes. This encompasses, but is not restricted to, data cleansing and normalization, dimensionality reduction, discretization, screening, and cleaning; thereafter, the researcher may utilize data segmentation and cross-validation to evaluate the model's generalization performance. Initially, data cleaning and normalization eliminate noise, enhance data quality, and guarantee the precision of subsequent studies. Zhang et al. (2021) eliminated duplicates from data media, translated non-English-language and nonalphabetic characters, and deactivated words to enhance text quality. Jebbor et al. (2022) also eliminated duplicate data points to mitigate data redundancy. In contrast to deleting duplicate data, Jiang et al. (2021) aggregated multiple data sources. It eliminated records with missing values to guarantee data integrity and precision, hence preventing model bias stemming from absent data.

Furthermore, certain research studies employed data decomposition or dimensionality reduction approaches to extract pertinent critical information and minimize data volume. Li et al. (2023) and He et al. (2022) employed the empirical mode decomposition (EMD) and complementary ensemble empirical mode decomposition (CEEMD) methodologies to deconstruct intricate time series data and extract components with varying frequencies, effectively addressing the nonlinear and nonsmooth challenges present in the data. By decomposing the original data into many intrinsic modal functions using the CEEMD technique, the researcher can identify the primary trend information in the low-frequency components while attenuating the noise predominantly present in the high-frequency region. Identifying probable themes from extensive data is challenging, and the latent Dirichlet allocation (LDA) technique was employed to analyze SOS messages from citizens' social media during disasters to categorize topics and ascertain urgent demands (Gao et al. 2022). Similarly, to streamline data attributes and reveal insights, the study delineated an attribute simplification model grounded in rough set theory, which refined the attributes

Table 4. Emergency demand forecasting projects

Author	Disaster type	Country	Time span	Type of input
Li et al. (2022b)	ND	China	50 years, 2008	Geology, impact, demographic, simulation
Song et al. (2023)	PHE	China	2020–2021	Geology, medicine, meteorology, media, need (almost all)
Ramgopal et al. (2021)	GD	US	2015–2019	Time, meteorology
Jiang et al. (2021)	Others	US	2017–2018	Demographic, survey, simulation
Hao et al. (2023)	Others	China	The past 3 years	Event, impact, meteorology, infrastructure
Nguyen et al. (2022)	ND	US	2012, 2017	Media, meteorology, impact
Oktarina et al. (2020)	ND	Indonesia	2000–2019	Demographic, geology, event
Hu et al. (2021a)	GD	China	—	Meteorology, geology, infrastructure, index
Zheng et al. (2022)	ND	China	2018	Event, geology, demographic, socioeconomic, meteorology
Jin et al. (2021)	GD	Japan	—	Medicine, demographic, socioeconomic
Shao et al. (2020)	GD	China	2008, 2012	Event, impact, socioeconomic, index
Li et al. (2023)	PHE	China	2014–2018	Medicine, index, demographic
Fang et al. (2021)	GD	China	2008, 2010	Geology, impact, medicine, demographic, need
He et al. (2022)	PHE	China	2012–2021	Socioeconomic, medicine
Nguyen et al. (2023)	GD	France	2017, 2020	Media, need
Barros et al. (2021)	PHE	Chile	8.5–10 year span	Time, need, model given parameter
Tian and Mei (2023)	GD	China	2022	Demographic, model given parameter
Wang et al. (2022)	PHE	China	2020	Demographic, model given parameter
Li et al. (2022a)	GD	US	2011–2012	Medicine, impact, infrastructure, meteorology, model given parameter
Edejer et al. (2020)	PHE	World	2020	Demographic, medicine, model given parameter, rescue, index
Chen et al. (2022)	ND	China	2013–2019	Demographic, need, hydrology, impact, media, model given parameter, index
Marzook et al. (2023)	GD	Sri Lanka	—	Demographic, medicine, survey
La et al. (2022a)	ND	China	2004–2019	Media, infrastructure, rescue, impact
Toland et al. (2023)	ND	US	Up to 2023	Demographic, infrastructure, impact, index
Dachyar and Ghifari (2020)	ND	Indonesia	2006–2019	Geology, impact, infrastructure, time
Li and Liu (2023)	PHE	—	2018–2022	Event, medicine, meteorology, index, impact
Gao et al. (2022)	ND	China	2021	Media, socioeconomic, geology, infrastructure, demographic, index, impact
Mallouhy et al. (2022)	GD	France	2015–2021	Event, infrastructure, index
Liang et al. (2024)	ND	Multiple	—	Geology, need, event, impact, socioeconomic, infrastructure, media
Zhu et al. (2022)	ND	Japan	2011	Need, demographic, model given parameter
Wang and Jia (2023)	PHE	China	2022	Medicine
Chen and Ji (2022)	PHE	US	2021	Media, demographic, survey, geology, socioeconomic
Zhang et al. (2021)	ND	Multiple	2013–2016	Media, demographic, model given parameter
Kamar et al. (2021)	PHE	Lebanon	2020	Medicine, demographic
Geng et al. (2024)	ND	China	2008	Impact, model given parameter, demographic, index, simulation
Biswas et al. (2024)	ND	Turkey	2015–2023	Event, demographic, socioeconomic, impact, infrastructure, geology, index
La et al. (2022a)	ND	China	2004–2019	Event, demographic, impact, rescue, infrastructure
Ma et al. (2023)	PHE	China	2022	Medicine, impact, infrastructure, index
Hu et al. (2021b)	ND	China	—	Event, demographic, impact
Jebbor et al. (2022)	GD	—	—	Medicine, demographic, impact, model given parameter, event
Huang et al. (2022)	ND	China	—	Media, index, infrastructure, socioeconomic, demographic
Fei and Wang (2022)	ND	China	2008–2018	Event, impact, need, demographic, socioeconomic, infrastructure
Chen and Ji (2021)	ND	US	2017	Media, survey, demographic, time
Jiang et al. (2024)	PHE	China	2021	Demographic, medicine, model given parameter, socioeconomic
Hezam (2022)	PHE	Multiple	2019–2020	Medicine, demographic, socioeconomic, index
Lin et al. (2020)	GD	China	2016–2018	Demographic, event, need, time, socioeconomic, infrastructure
Guo et al. (2021)	GD	—	—	Impact, rescue, index, infrastructure, model given parameter
Caro et al. (2021)	PHE	Multiple	2020	Medicine, event, simulation
Cheng et al. (2021)	PHE	Multiple	2019–2020	Medicine, index, time

of the original data utilized for the earthquake scenario and the desktop simulation dataset (Li et al. 2022a).

Conversely, if the quantity or nature of data gathered in the present study is inadequate, data integration and augmentation represent effective solutions; the research conducted by Nguyen et al. (2023) amalgamated structured data on drug demand with unstructured data sourced from news articles, thereby integrating and augmenting the data to improve the model's generalization capacity. This approach closely resembles multimodal fusion. Multimodal fusion emphasizes the processing and analysis of data from several modalities to accomplish integrated applications and more intricate jobs.

The discretization of continuous data has been partially explored for practical or algorithmic purposes (Li et al. 2022a; Gao et al. 2022). This process allows for the transformation of continuous attributes based on inherent rules. Another crucial aspect of data preprocessing is normalization, which standardizes data of varying scales to a uniform range, mitigating the impact of scale discrepancies, enhancing model performance, and expediting convergence (Song et al. 2023; Chen et al. 2022).

Ultimately, due to the variety of input data types for emergency demand forecasting, processing techniques are essential for enhancing outcomes. The enhanced ant colony algorithm optimizes the weights and thresholds in a back propagation (BP) neural network, thereby augmenting the network's convergence speed and predictive accuracy (Chen et al. 2022). Additionally, certain studies have employed data segmentation and cross-validation to evaluate

Table 5. Index and indicator involved in the review

Index	Content	Purpose
Equipment index	Equipment technical status, operational defects, and failure rate	Indicates the vulnerability of the equipment
Terrain index	Terrain undulation, elevation, curvature	Describes terrain complexity
Apparent temperature	(Daytime or night) and (low or high)	Evaluates material requirements affected by atmospheric conditions
Reserve demand of medicines	Cases × demand coefficient	Forecasts emergency drug reserves
Oxford stringency index	Combined measure derived from nine different response metrics	Measures the level of COVID-19 mitigation measures
Circulation path evaluation index	A comprehensive evaluation of the pros and cons of the circulation path in the disaster area	Provides a basis for calculating and modeling
Predisaster food insecurity index	Aggregating and weighting these sociodemographic variables appropriately	Models community-level risk disparities in both resource access and food security
Risk level of Twitter	Risk expression level of tweet text, flood vulnerability of tweet location	Shows the possibility of disaster occurrence
Social vulnerability index	Combining various sociodemographic variables into a single composite score	Measures social vulnerability
Infrastructure disruption index	Power, water, transportation system damage	Simulates the ShakeOut scenario
Location hazard index	Identify clusters, determine factors, calculate cluster hazard index, adjust for surrounding conditions	Measures the potential hazard of a location
Resilience-level evaluation index system	Economic, social, environmental, infrastructural, institutional resilience	Provides a comprehensive assessment of the resilience level of megacities
Forest fire danger index	A mathematical formula	Estimates the potential for large fire ignitions in forested regions
Materials demand satisfaction rates	A mathematical formula	Evaluates the effectiveness of disaster response strategies
Shortage of medical and psychological services	A mathematical formula	Estimates the shortage of medical and psychological services
Degree of intolerance to pain	A mathematical formula	Measures victims' pain tolerance
Evaluation index system of disaster relief demand	Emergency support demands, emergency rescue demands, basic life support demands and public infrastructure support demands	Identifies specific disaster relief needs to improve the timeliness and accuracy of emergency rescue
Human development index	Health, education, standard of living	Understanding well-being and development beyond income alone
Legatum prosperity index	Based on 12 pillars	Measures and understands how prosperity is created and sustained across different countries
Importance index	Average daily cargo flow, link irreplaceability, mean time to repair	Quantifies the importance of links (or nodes) within a multimodal transport network
ED crowding scores	ED overcrowding study, the Emergency Department Work Index Score, occupancy rate	Detects and quantifies crowding
Emergency severity index	ESI five levels	Classifies patients in the emergency department and determines priorities according to the severity and urgency of the patient's condition

Note: ED = emergency department; and ESI = emergency severity index.

the model's generalization capacity, mitigate overfitting, and enhance model robustness (Li et al. 2022a). In summary, data preprocessing is an essential phase in the research process of emergency demand forecasting, significantly influencing the accuracy and reliability of the forecasting model.

Modeling

Emergency demand forecasting is a multifaceted process that entails anticipating the types and quantities of demand following a disaster while also accounting for numerous uncertainties to assure the validity and reliability of the outcomes in emergency response (Vollmer et al. 2021; Yang et al. 2021). To fulfill these forecasting requirements, emergency management must depend on advanced modeling techniques, robust data collection and processing capabilities, and broad community acceptance to minimize losses during emergencies effectively. The modeling techniques documented in the literature can be categorized into six groups: ML, statistical and mathematical models (SMM), NLP, optimization and decision models (ODM), forecasting tools and simulation (FTS), and data integration and conceptual frameworks (DICF). Table 5 and Fig. 5 illustrate the modeling methodologies employed in each investigation. Table 6 summarizes the forecasting techniques adopted in the reviewed studies. Subsequently,the advantages, limitations, and application scenarios of the six forecasting techniques are summarized in Table 7.

Machine Learning

Machine learning is becoming a crucial instrument in investigating emergency demand forecasting. It provides numerous forecasting benefits via automated feature extraction, intricate and multidimensional data processing, and dynamic state updating. Fig. 5 enumerates six categories of machine learning utilized in the literature. Neural networks and deep learning are interrelated methodologies, with neural networks serving as a broad term for any hierarchical computing model comprising neurons and weights. Deep learning is a specific subset of learning this word, denoting neural networks with numerous hidden layers, emphasizing the deep architecture of the neural network (Oktarina et al. 2020; He et al. 2022; Jebbor et al. 2022).

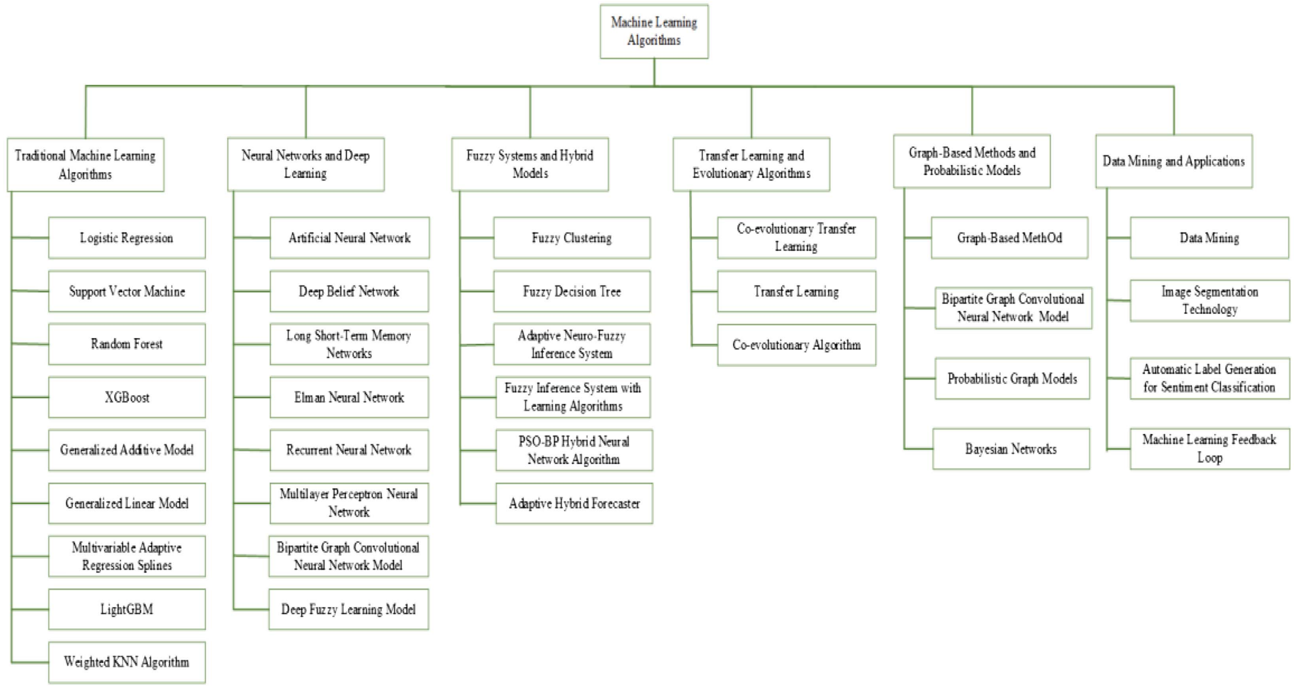

Fig. 5. Taxonomy of emergency demand prediction method (ML).

Moreover, numerous network topologies documented in the literature possess distinct benefits and applications pertinent to their specific contexts (Nguyen et al. 2022; Hu et al. 2021a; Zheng et al. 2022; Jin et al. 2021; Li et al. 2023; He et al. 2022; Jebbor et al. 2022; Lin et al. 2020). Current research encompasses not just prevalent neural network models and architectures but also fuzzy systems, conformance models, transfer learning, co-evolutionary algorithms, image-based and probabilistic models, data mining, and many other applications. This paper classifies certain fundamental and prevalent models as conventional models for subsequent classification and comparison. Migration learning significantly diminishes reliance on extensive labeled datasets and enhances the precision of contingency predictions in scenarios with limited data or analogous tasks (Song et al. 2023; Zheng et al. 2022).

Statistical and Mathematical Models

Statistical and mathematical models characterize, examine, and forecast global phenomena utilizing data formats and statistical techniques. These models elucidate the intrinsic relationships among data and facilitate informed decision-making. These models can exist in either deterministic or probabilistic formats. They may concurrently manage several kinds of data, including time series, categorical data, and continuous data. The principal SMM models discussed in the literature include time-series models, rough sets, entropy and information theory, gray prediction, Bayesian approaches, and fuzzy learning (He et al. 2022; Wang et al. 2022; Li and Liu 2023; Chen and Ji 2022; Huang et al. 2022; Cheng et al. 2021). Time-series models are crucial for emergency demand forecasting initiatives. Diverse time-series models offer forecasting versatility and aid in addressing intricate scientific challenges. ARIMA and seasonal autoregressive integrated moving average with exogenous variables (SARIMAX) can address trends and seasonal fluctuations in time series while accounting for exogenous elements (Li et al. 2023; He et al. 2022), whereas a vector autoregressive with exogenous (VARX) integrates exogenous variables for forecasting in more intricate scenarios and excels in multivariate analysis (Nguyen et al. 2023). Moreover, Grey model (GM) (1,1) and metabolic gray models are appropriate for scenarios with limited data and can provide more reliable projections of emergency demand (Wang and Jia 2023; Ma et al. 2023).

Natural Language Processing

In the information era, NLP technology can facilitate information extraction and processing, augment the efficacy of predictive models, enable real-time crisis monitoring, and enhance the interpretability of predictive models. In emergencies, the LDA model is a topic model that discerns primary topics and associated terminology within a corpus of texts, particularly in identifying distress SOS messages (Gao et al. 2022). Additionally, the biterm topic model, an enhancement of the LDA model, is tailored for the topic modeling of brief texts to ascertain emergency demand subjects by capturing co-occurring lexicon (Zhang et al. 2021). Furthermore, analyzing user behavior, sentiment trends, and topic developments on social media during disasters can assist decision makers in recognizing potential crises and mitigating rescue obstacles (Nguyen et al. 2023; Chen and Ji 2021).

Optimization and Decision Models

This category encompasses several optimization and decision-making techniques that utilize scientific and mathematical methodologies to enhance the allocation of emergency resources, minimize costs and risks, and maximize the effectiveness of emergency response. The literature mainly addresses inventory management and supply chain optimization (Tian and Mei 2023; Hu et al. 2021b), as well as decision support and multiobjective optimization (Dachyar and Ghifari 2020; Hu et al. 2021b Fei and Wang 2022; La et al. 2022b; Geng et al. 2024); resource allocation and location optimization (Wang et al. 2022; Geng et al. 2024); logistics and path optimization (Shao et al. 2020); uncertainty management

Nat. Hazards Rev., 2026, 27(1): 03125003

Table 6. Emergency demand forecasting by technique

Author	ML	SMM	NLP	ODM	FTS	DICF
Li et al. (2022b)	X	X	—	—	X	—
Song et al. (2023)	X	—	—	—	—	—
Ramgopal et al. (2021)	X	X	—	—	—	—
Jiang et al. (2021)	X	X	—	—	X	X
Hao et al. (2023)	X	X	—	—	—	—
Nguyen et al. (2022)	X	X	X	—	—	—
Oktarina et al. (2020)	X	—	—	—	—	—
Hu et al. (2021a)	X	X	—	—	X	—
Zheng et al. (2022)	X	X	—	—	—	X
Jin et al. (2021)	X	—	—	X	—	—
Shao et al. (2020)	X	X	—	—	—	X
Li et al. (2023)	X	X	—	—	—	—
Fang et al. (2021)	—	X	—	—	X	X
He et al. (2022)	X	X	—	—	—	—
Nguyen et al. (2023)	X	X	X	—	—	X
Barros et al. (2021)	X	—	—	X	X	X
Tian and Mei (2023)	—	X	—	X	—	—
Wang et al. (2022)	—	X	—	X	—	—
Li et al. (2022a)	—	X	—	—	—	—
Edejer et al. (2020)	—	X	—	—	X	—
Chen et al. (2022)	X	X	—	X	X	—
Marzook et al. (2023)	X	—	—	—	X	X
La et al. (2022a)	—	X	—	X	—	X
Toland et al. (2023)	—	X	—	X	X	X
Dachyar and Ghifari (2020)	—	X	—	—	X	—
Li and Liu (2023)	—	X	—	X	—	—
Gao et al. (2022)	X	—	X	—	—	—
Mallouhy et al. (2022)	X	—	—	—	X	—
Liang et al. (2024)	X	X	—	—	—	X
Zhu et al. (2022)	—	X	—	—	X	—
Wang and Jia (2023)	—	X	—	—	—	—
Chen and Ji (2022)	—	X	X	—	X	—
Zhang et al. (2021)	—	—	X	—	—	X
Kamar et al. (2021)	—	—	—	—	X	—
Geng et al. (2024)	X	X	—	X	X	—
Biswas et al. (2024)	X	—	—	—	—	X
La et al. (2022a)	—	X	—	—	—	—
Ma et al. (2023)	—	X	—	X	—	—
Hu et al. (2021b)	X	X	—	—	—	—
Jebbor et al. (2022)	X	—	—	—	—	X
Huang et al. (2022)	—	X	X	X	—	X
Fei and Wang (2022)	—	X	—	—	—	X
Chen and Ji (2021)	—	X	X	—	—	X
Jiang et al. (2024)	—	X	—	X	—	—
Hezam (2022)	—	X	—	X	—	—
Lin et al. (2020)	X	X	—	—	X	X
Guo et al. (2021)	—	X	—	X	—	—
Caro et al. (2021)	—	X	—	—	X	—
Cheng et al. (2021)	—	X	—	—	—	—

(Tian and Mei 2023); risk assessment (Toland et al. 2023); prioritization (Liang et al. 2024); and additional topics. The optimization and decision-making models have significantly improved the efficacy and responsiveness of emergency management in demand forecasting during crises. In the study conducted by Geng et al. (2024), a three-tier mathematical model for emergency rescue was proposed to address the urgent need for rapid response. Additionally, a hybrid algorithm was developed to solve the model by examining various types of emergency shelters concerning siting, evacuee allocation, and on-demand allocation dimensions.

Forecasting Tools and Simulation

FTS comprises a compilation of developed and constrained predictive instruments and simulation frameworks. Each of these tools and simulation systems possesses distinct application scenarios and advantages; for instance, time-share simulation (Li et al. 2022a) and Monte Carlo methods (Jiang et al. 2021) are predominantly oriented toward multioption evaluation under dynamic uncertainty. In contrast, system dynamics (Chen et al. 2022) and CEEMMD (He et al. 2022) are more adept at addressing complex systems and data. In summary, these strategies enhance the efficacy of emergency response and significantly mitigate the danger of resource scarcity.

Data Integration and Conceptual Frameworks

DICF is a methodical strategy designed to enhance emergency responses by augmenting the precision and dependability of predictions via the effective amalgamation of various data types (structured, unstructured, and semistructured data) (Fang et al. 2021; Nguyen et al. 2023; Li et al. 2022b; Lin et al. 2020). Significant data sources identified in the research by Lin et al. (2020) include web mapping services, social media, and crowdsourcing systems, which facilitate real-time insights into dynamic population distribution and disaster-affected regions, thereby enhancing the accuracy of emergency needs predictions in real time.

Evaluation

According to the output results, all emergency demand forecasting issues may be efficiently classified, with the specific forecasting output, performance metrics, and performance detailed in Table 8. Upon categorizing the diverse output data, it is evident that the current research encompasses a range of crisis scenarios, demand categories, and response strategies, highlighting the intricacy of forecasting multiple demand kinds, including resources, personnel,

Table 7. Comparative analysis of forecasting method

Method	Advantage	Limitation	Scenario
ML	Integrates with other data or methods; supports dynamic decisions	The limitation of data acquisition	Earthquakes, floods, power outages, epidemics, firefighting, among others
SMM	Easy to apply; handles uncertainty; adapts to small sample size	Math-heavy; challenge for big data; sampling bias; does not adapt to fast changes	Epidemics, typhoons, hotel demand, among others
NLP	Handles unstructured text; detects emotions; supports Volunteered Geographic Information (VGI) use	Sparse/noisy data; social media bias; demographic bias	Epidemics, disasters, scandals, natural environments, among others
ODM	Intuitive; interpretable; sensitivity analysis; aids decision-making	Relies heavily on accurate data; Np-hard problem; challenge for unstructured data	Not limited to specific scenario
FTS	Simple; user-friendly (predictive tools); handles complexity (simulation methods)	No effect in complex cases (predictive tools); high computational needs (simulation)	Natural disasters, among others
DICF	Supports data fusion; adapts to complex, multisystem, high-volume environments	Technical threshold	Not limited to specific scenario

Note: Np-hard = nondeterministic polynomial-time hard.

Table 8. Performance comparison of the forecast models

Author	Output	Performance measurement	Performance
Li et al. (2022b)	Cumulative number of deaths, material demand	R^2, MSE, general mean prediction error	0.9819 (final round)/0.0519 (final round)/10.96%
Song et al. (2023)	Demand forecasts for 10 types of COVID-19 prevention and control materials	Prediction accuracy, success rate, demand satisfaction rate	Around 80%/83.33% (mask)/vagueness
Ramgopal et al. (2021)	Hourly EMS dispatches	R^2, MSE, MAE, RMSE	0.92/217.00/11.30/14.7
Jiang et al. (2021)	Disaster preparedness prediction	Accuracy, specificity, F1 score, AUC	0.656/0.757/0.734/70.9% (all SVM)
Hao et al. (2023)	Demand forecasting for rush repair spare parts	Precision, RMSE	0.96/12.38 (all seven classes)
Nguyen et al. (2022)	People's needs during hurricane events	SMC, accuracy, precision, recall, F1 score	0.882 (Irma)/0.932/0.934/ 0.939/0.937
Oktarina et al. (2020)	Number of damaged buildings and casualties due to earthquakes	MSE	520,924
Hu et al. (2021a)	The demand for emergency material resources in the power grid under meteorological disasters	Accuracy, Pearson correlation coefficient	85.52% (emergency repair personnel)/corrected wind value (0.823)
Zheng et al. (2022)	The demand for disaster relief supplies (175 types)	Average regression error, success rate	1.74/72.37%
Jin et al. (2021)	Demand for emergency medical services	Accuracy, F1 score, precision, recall	77.3%–87.7%/80.4%–89.5%/—/—
Li et al. (2023)	Emergency medicine reserve demand	MAPE	3.87%
He et al. (2022)	Hotel demand in the context of public health emergencies	MAE, RMSE, MAPE	24,820.78 /28,606.31/ 5.63%
Nguyen et al. (2023)	The demand for pharmaceutical products during times of disruption	Global accuracy, sentiment score accuracy, mean error, MAE, MAPE, RMSE (demand deviation)	83%/86%/3.51/15.48/ 3.90%/25.60
Barros et al. (2021)	Demand for emergency medical services and corresponding hospital capacity management	MAPE, MSE	5.61%/145,861 (support vector regression–medical demand)
Wang et al. (2022)	The demand for emergency medical facilities	Minimum ratio, maximum ratio, mean relative error, MSE ratio, small error probability, relevancy degree accuracy	0.7645/0.8254/0.009/ 0.054/0.993/1
Chen et al. (2022)	The demand for emergency supplies during flood disasters	Average deviation rate, best solution rate, MSE, MAE, SSE	—, —, 0.194/0.335/2.981
La et al. (2022b)	Demand for different types of rescue personnel required for specific tasks during a geological disaster	Relative error	4.26%
Dachyar and Ghifari (2020)	The demand for various emergency logistic supplies in the aftermath of an earthquake	MAPE	10.48% (food)
Li and Liu (2023)	Daily necessities, test materials, and the number of confirmed cases at various times	MAE, MAPE, RMSE, posterior variance test, residual test ($\alpha = 0.05$), correlation test ($\rho = 0.5$)	$618/0.18/0.08/C = 0.46$, $p = 0.81/0.034/0.71$ (All City 1)
Gao et al. (2022)	The areas needing evacuation and the resources required	Accuracy, precision, recall, F1 score, AUC	0.7823/0.7945/0.9018/ 0.8448/0.8115
Zhu et al. (2022)	The demand for critical food supplies in the aftermath of the 2011 Tohoku earthquake	R^2, F-statistic tests	57%/—
Wang and Jia (2023)	The predicted number of asymptomatic infected cases and the corresponding demand for emergency supplies	MAPE	3.277%
Chen and Ji (2022)	The actual public demand during disasters	RMSE, the normalized testing errors	—/ < 5%
Zhang et al. (2021)	The distribution of the relief supplies needs	Accuracy, precision, recall, F1 score	0.68/0.64/0.76/0.6
Biswas et al. (2024)	Death, injuries, infrastructure damages, and cascading disaster/relief supply	Accuracy rate, MAE, MSE, RMSE, R^2, RMSLE, MAPE	88.99%/0.0257 /0.0022/ 0.0303 /0.5845/0.0191 /0.1784 (Decision Tree)
Ma et al. (2023)	The number of infections and material demand	MAPE	6.7050%
Hu et al. (2021b)	The demand of emergency material and the number of post-earthquake casualties	Error	11.3% (death)
Jebbor et al. (2022)	The consumption amounts of 12 assets during a burn mass casualty incident	MSE, accuracy percentage	0.0125901/97.70% (ANFIS models albumin testing)
Fei and Wang (2022)	The demand for emergency materials in disaster areas	Accuracy rate	84.89%
Hezam (2022)	Final epidemic size in various high-priority countries	RMSE	294.893
Lin et al. (2020)	The demand for relief supplies during urban flood disasters	Correlation coefficients	0.923 (blankets)
Cheng et al. (2021)	The hourly occupancy of the emergency department	MSE, MAE, MAPE	16.196/3.158/0.050 (1-h-ahead prediction)

Note: EMS = emergency medical services; AUC = area under the curve; SMC = the simple matching coefficient; SSE = sum of squared errors; and RMSLE = root mean squared logarithmic error. All City 1 refers to the aggregation of all evaluated indicators corresponding to the anonymized case study labeled as City 1.

and medical services. It also elucidates the intricacies of predicting many demand categories, including supplies, staffing, and medical services during emergencies, and the associated research findings with practical relevance might assist decision makers in pre-emptively preparing for crises.

A literature review indicates that most findings from these studies are regression issues rather than categorization issues, including metrics such as mortality rates, hospital department occupancy rates, and other predictive outcomes. Table 8 delineates the performance metrics of the entire research endeavor, and the unreferenced material has been excluded from the table. The table analysis indicates that the assessment metrics, arranged in descending order of frequency, are mean absolute percentage error (MAPE), mean square error (MSE), accuracy, mean absolute error (MAE), root-mean-square error (RMSE), and F1 score. Initially, MAPE quantifies the discrepancy between expected and actual values, expressed as a percentage, enabling data comparison across varying scales. Nevertheless, values approaching zero exacerbate the mistake and influence the outcomes.

Conversely, MSE quantifies the squared mean of the discrepancy between expected and actual values, exhibiting high sensitivity to substantial errors. Second, accuracy is a frequently employed evaluation parameter in classification models (Jiang et al. 2021; Nguyen et al. 2022). However, its definitions differ across the literature. In Song et al. (2023), accuracy denotes the discrepancy between actual and expected demand for each material type, which is subsequently aggregated to derive the overall deviation. In the study of Nguyen et al. (2023), global accuracy and sentiment score accuracy denote the comprehensive evaluation of the sentiment analysis model and its efficacy in classifying positive and negative sentiments in news headlines, respectively. Each labeled accuracy possesses its distinct formula. Moreover, like MAE, RMSE quantifies the disparity between the predicted values and the actual data. At the same time, the F1 score serves as a balanced, holistic assessment of the classification model, integrating the equilibrium between precision and recall.

Overall Analysis

Fig. 6 illustrates that the primary categories of input data for predicting diverse emergency requirements include demographic, impact, and socioeconomic factors. This type of data offers a more pertinent representation of a community's or region's vulnerability and needs in emergency response practices; thus, information such as population distribution, disaster impact, and household financial status can be employed to enhance the precision and promptness of emergency demand forecasts.

Fig. 7 illustrates the application of technologies in emergency demand forecasting in recent years. The utilization of ML, SMM,

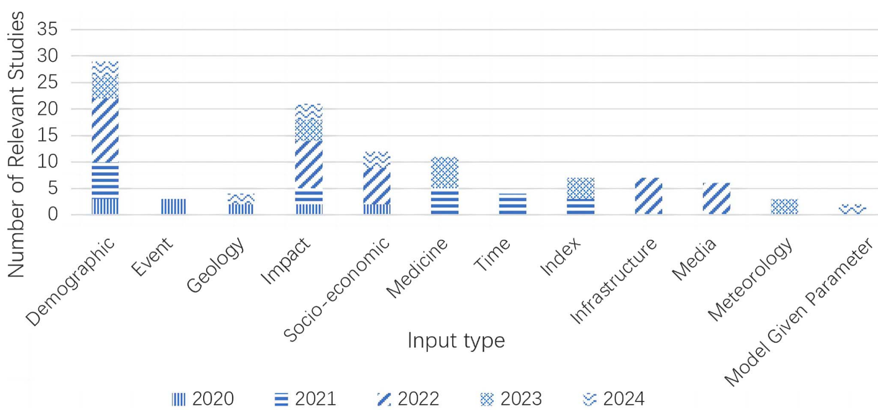

Fig. 6. Distribution of input type over the past 5 years.

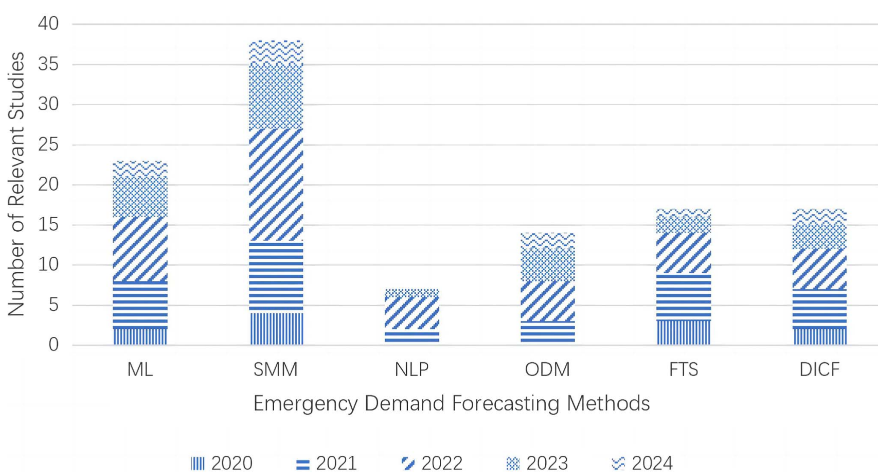

Fig. 7. Distribution of emergency demand forecasting methods over the past 5 years.

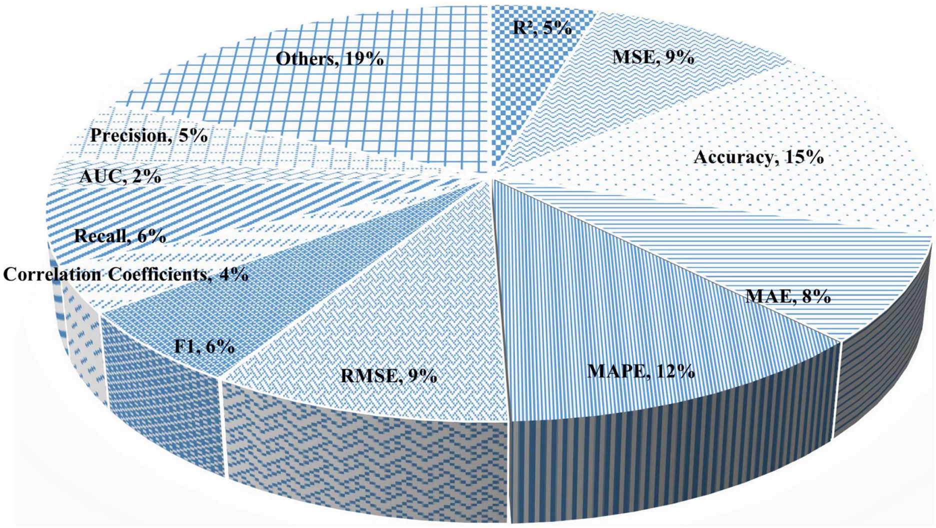

Fig. 8. Usage proportion of different evaluation metrics.

and DICF is predominant, but the application of NLP, FTS, and ODM is comparatively minimal. Moreover, ML and SMM methodologies have exhibited relative stability in the volume of technical reports over recent years while demonstrating progressive growth over time. In emergency prediction, ML and statistical and mathematical models (SMM) have consistently been the predominant strategies, garnering increasing attention over time relative to other methodologies, while research in data fusion and conceptual framework categories is also swiftly advancing.

Fig. 8 illustrates the research effort's performance metrics, prominently featuring MAPE, accuracy, MSE, and RMSE alongside other represented performance parameters. The variety of these indicators illustrates the complex requirements of model evaluation, including the performance assessment of categorical models and the accuracy evaluation of regression models.

Fig. 9 illustrates the annual MAPE performance. Fig. 8 clearly indicates the existence of numerous performance measuring categories. Ten studies referencing MAPE were retrieved from the literature library, revealing isolated study works in certain years, and a maximum of five research works in 2023, with the corresponding experimental results illustrated in a line graph. The experimental findings indicate that numerous investigations yield a MAPE ranging from 0% to 7%, whereas a limited number of trials attain 18%. These findings demonstrate the precision and intricacy of contingency demand forecasting.

Ultimately, the reviewed literature predominantly focuses on underdeveloped nations, while disaster assistance in wealthy countries represents a prospective avenue for future research. The analysis of the input data type reveals that current research on emergency resource prediction emphasizes the significance of multisource

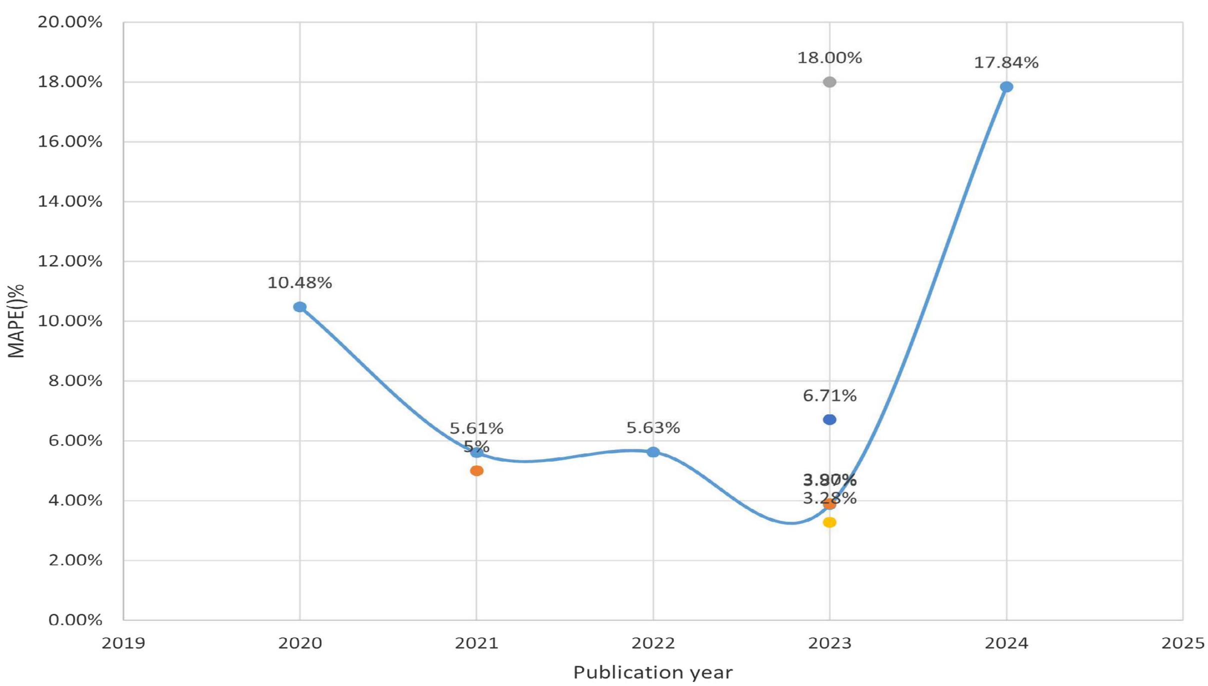

Fig. 9. Distribution of MAPE over the past 5 years.

data, encompassing various indications. Furthermore, the studies of the gathered literature identified six primary categories of technologies, and the complete process of emergency demand forecasting was delineated.

Future Research Directions

The current research indicates that fuzzy logic is extensively employed in emergency demand forecasting, particularly in addressing data uncertainty and ambiguity. Triangular fuzzy numbers can efficiently quantify and convert fuzzy indicators for enhanced evaluation accuracy (Huang et al. 2022). A hierarchical clustering approach utilizing fuzzy logic has been proposed to dynamically organize emergency shelters and classify them according to their assistance degree, hence elucidating the rescue strategy (Geng et al. 2024). The fundamental principle of the preceding two methods is to extract essential information through assessment; conversely, some studies integrate fuzzy logic with neural networks to employ fuzzy rules and adaptive mechanisms for managing nonlinear and complex data, thereby improving the efficacy of emergency demand forecasting (Jebbor et al. 2022). Fuzzy inference systems employ fuzzy logic and may generate precise predictions in intricate and uncertain contexts, rendering them appropriate for use in highly unpredictable scenarios. Prospectively, there is significant potential for using fuzzy logic in emergency demand forecasting, particularly among escalating uncertainty. Two potential developmental trajectories can be suggested: first, deep neural network technology may be employed to create an emotion analysis system for disaster victims to monitor their emotional fluctuations during disaster relief efforts. When integrated with fuzzy decision theory, this information can serve as an objective criterion for evaluating the necessity of disaster relief. A hierarchical system comprising low-dimensional fuzzy systems might mitigate the fuzzy rule explosion phenomenon by amalgamating inference rules, hence improving the application of predictions in a big data context.

This paper has found other phenomena; whereas numerous publications address algorithmic approaches for emergency demand forecasting, most emphasize model resolution rather than explicitly quantifying material needs. For instance, certain research employed a gray prediction model to assess the cumulative case count during an outbreak, subsequently inferring the emergency requirements and ultimately applying a genetic algorithm to address the issue of medical facility placement (Wang et al. 2022). Conversely, the improved ant colony optimization-back propagation neural network (IACO-BP) algorithm excels at forecasting emergency material requirements during flood disasters, enhancing rescue operations' speed and precision (Chen et al. 2022). Upon comparing the studies on the algorithms in these two bodies of literature, it is evident that the latter is more direct and significantly contributes to the forecast of emergency material demand. In contrast to the constraints of conventional models in addressing uncertain data, future research may develop integrated interval prediction algorithms by incorporating metaheuristic algorithms alongside a synthesis of interval and individual predictions to enhance predictive accuracy. Simultaneously, metaheuristic methods can be integrated with machine learning techniques like XGBoost to improve the efficacy of emergency material need forecasts. Furthermore, from the standpoint of algorithmic development, future research may focus on the following aspects: performing sensitivity analysis to investigate the algorithms' responsiveness to variations in input parameters to improve robustness and incorporating probabilistic or Bayesian methods to quantify uncertainty, thereby facilitating comprehensive evaluation of predictive capability.

Furthermore, limited research has been identified that employs large language models (LLMs) for forecasting emergency demand. A large language model is a machine learning algorithm designed to identify and forecast linguistic patterns by analyzing extensive text, code, literature, and book content from diverse digital sources; its training can be supervised, initially relying on human evaluations, and may evolve into self-supervised learning. Large models in emergency management can autonomously analyze knowledge graphs, thus improving the efficiency of swift decision-making (Chen et al. 2024). In the future, large language models should be extensively included in emergency management, integrating with knowledge graph technology to direct the reasoning of these models using prompt engineering, ensuring effective question input and dependable response output. The research on prompt engineering is crucial for enhancing the consistency and reliability of LLM modeling. Researchers must provide precise descriptions of emergency demand prediction scenarios, creatively devise various prompts to direct LLMs to employ diverse cognitive approaches in their responses, and evaluate their efficacy in intricate tasks, thereby enhancing the model's robustness.

Ultimately, comparative studies are deficient concerning coupled, secondary, or derived dangers. Emergency events typically initiate a subsequent series of occurrences, resulting in a dynamically evolving state over time; when the impact of the initial emergency has not yet diminished, a derivative event may arise abruptly, presenting a research challenge in demand forecasting amid such interconnected disasters. The research approach revealed numerous intriguing aspects of emergency management that can provide insights from existing studies. The investigation of dynamic truck and drone synergy strategies has been implemented in the material dispatching process within a city (Long et al. 2023), necessitating a stronger connection to emergency response requirements to enhance urban emergency resilience. Additionally, factors such as the reliability of predictions (Kholaif et al. 2024) and the satisfaction of catastrophe victims (Davies and Alaabi 2023) may also be emphasized in emergency forecasting.

Conclusion

This paper presents a contemporary survey of the literature on emergency demand forecasting. It concentrates on analyzing predictive regression issues from 2020 to 2024. This analysis meticulously examines 49 studies, categorizing them according to the inputs and outputs utilized and the forecasting methodologies applied.

Demographic information is the predominant source utilized for emergency demand forecasts instead of technical indications. Furthermore, the findings from the prior analysis indicate that numerous research endeavors incorporate various kinds of data simultaneously. Social network data enhance the precision of emergency response prediction models; examining sentiment states and trends on social networks and self-media platforms facilitates a more thorough and accurate comprehension of current emergency response dynamics.

The scientific community is increasingly utilizing sophisticated machine learning modeling methods. An illustration is the bipartite graph convolutional neural network (GCN) model and multilayer perceptron (MLP) neural network; sophisticated and novel methods outperform conventional SVM and ANN. This paper's research delineates five significant sorts of methodologies, apart from machine learning, that are crucial for the advancement of emergency demand forecasting.

Consequently, future research endeavors should prioritize the extensive use of multisource data and the integration of diverse methodologies to enhance the efficacy of forecasting initiatives. NLP approaches can be utilized to evaluate network sentiment in crisis zones and gauge the demand for topics, resources, and services. This methodology provides a more lucid and precise comprehension of circumstances during a disaster or other emergencies. Due to the intricacies of real-world data, research in feature engineering is essential for optimizing the input size of predictive data, enhancing the comprehension of data by predictive models and augmenting the models' predictive capabilities.

This review has several restrictions. Due to Google Scholar's extensive search capabilities, only the 200 most pertinent articles were chosen for this work, potentially overlooking relevant publications present in the database. Second, while defining critical phrases requires substantial effort and numerous repetitions, the potential for overlooking infrequently used terms cannot be entirely eliminated. Due to the nature of this research, namely, an academic literature evaluation, much of the work conducted in emergency response activities or exercises may not be represented.

Data Availability Statement

Some or all data, models, or code that support the findings of this study are available from the corresponding author upon reasonable request, along with a list of all studies initially identified and reviewed as potential candidates but excluded for not meeting the inclusion criteria.

Acknowledgments

This study was funded by the Ministry of Education Industry-University Cooperation Collaborative Education Program (202002076027) and Heilongjiang Province university think tank open project ZKKF2022208. The authors express their gratitude to the reviewers and editors for their valuable feedback and suggestions, which helped enhance the quality of this paper.

Author Contributions

Baiqing Sun: Conceptualization; Funding acquisition; Supervision. Han Zhang: Formal analysis; Writing – original draft; Writing – review and editing. MinWei Deng: Data curation; Visualization. Shuai Liu: Visualization. Ruoshu Wang: Project administration.

References

Altay, N., and A. Narayanan. 2022. "Forecasting in humanitarian operations: Literature review and research needs." *Int. J. Forecasting* 38 (3): 1234–1244. https://doi.org/10.1016/j.ijforecast.2020.08.001.

Armstrong, J. S. 2001. *Principles of forecasting: A handbook for researchers and practitioners*. New York: Springer.

Barros, O., R. Weber, and C. Reveco. 2021. "Demand analysis and capacity management for hospital emergencies using advanced forecasting models and stochastic simulation." *Oper. Res. Perspect.* 8 (Jan): 100208. https://doi.org/10.1016/j.orp.2021.100208.

Biswas, S., D. Kumar, M. Hajiaghaei-Keshteli, and U. K. Bera. 2024. "An AI-based framework for earthquake relief demand forecasting: A case study in Türkiye." *Int. J. Disaster Risk Reduct.* 102 (Feb): 104287. https://doi.org/10.1016/j.ijdrr.2024.104287.

Caro, J. J., J. Möller, V. Santhirapala, H. Gill, J. Johnston, K. El-Boghdadly, R. Santhirapala, P. Kelly, and A. McGuire. 2021. "Predicting hospital resource use during COVID-19 surges: A simple but flexible discretely integrated condition event simulation of individual patient-hospital trajectories." *Value Health* 24 (11): 1570–1577. https://doi.org/10.1016/j.jval.2021.05.023.

Chambers, J., and K. Johnson. 1986. "Predicting demand for accident and emergency services." *J. Public Health* 8 (2): 93–103. https://doi.org/10.1093/oxfordjournals.pubmed.a043851.

Chee, M. L., et al. 2023. "Artificial intelligence and machine learning in prehospital emergency care: A scoping review." *iScience* 26 (8): 107407. https://doi.org/10.1016/j.isci.2023.107407.

Chen, F., J. Chen, and J. Liu. 2022. "Forecast of flood disaster emergency material demand based on IACO-BP algorithm." *Neural Comput. Appl.* 34 (5): 3537–3549. https://doi.org/10.1007/s00521-021-05883-1.

Chen, M., Z. Tao, W. Tang, T. Qin, R. Yang, and C. Zhu. 2024. "Enhancing emergency decision-making with knowledge graphs and large language models." *Int. J. Disaster Risk Reduct.* 113 (Oct): 104804. https://doi.org/10.1016/j.ijdrr.2024.104804.

Chen, Y., and W. Ji. 2021. "Public demand estimation following disasters through integrating social media and community demographics." In *Proc., 2021 Winter Simulation Conf. (WSC)*, 1–10. New York: IEEE. https://doi.org/10.1109/WSC52266.2021.9715440.

Chen, Y., and W. Ji. 2022. "Estimating public demand following disasters through Bayesian-based information integration." *Int. J. Disaster Risk Reduct.* 68 (Jan): 102713. https://doi.org/10.1016/j.ijdrr.2021.102713.

Cheng, Q., N. T. Argon, C. S. Evans, Y. Liu, T. F. Platts-Mills, and S. Ziya. 2021. "Forecasting emergency department hourly occupancy using time series analysis." *Am. J. Emerg. Med.* 48 (Oct): 177–182. https://doi.org/10.1016/j.ajem.2021.04.075.

Dachyar, M., and M. I. A. Ghifari. 2020. "A rapid decision model of disaster relief logistic, based on internet of things (IoT) data analytics and case-based reasoning." In *Proc., 2nd African Int. Conf. on Industrial Engineering and Operations Management (IEOM 2020)*, 473–483. Southfield, MI: Industrial Engineering and Operations Management Society International.

Davies, A., and F. Alaabi. 2023. "Responding to crisis and emergencies—Linking employee characteristics to levels of satisfaction with governance, procedures and mechanisms." *Int. J. Disaster Risk Reduct.* 96 (Oct): 103922. https://doi.org/10.1016/j.ijdrr.2023.103922.

Edejer, T. T. T., O. Hanssen, A. Mirelman, P. Verboom, G. Lolong, O. J. Watson, L. L. Boulanger, and A. Soucat. 2020. "Projected health-care resource needs for an effective response to COVID-19 in 73 low-income and middle-income countries: A modelling study." *Lancet Global Health* 8 (11): E1372–E1379. https://doi.org/10.1016/S2214-109X(20)30383-1.

Elbanna, A., D. Bunker, L. Levine, and A. Sleigh. 2019. "Emergency management in the changing world of social media: Framing the research agenda with the stakeholders through engaged scholarship." *Int. J. Inf. Manage.* 47 (Aug): 112–120. https://doi.org/10.1016/j.ijinfomgt.2019.01.011.

El-Bouri, R., T. Taylor, A. Youssef, T. Zhu, and D. A. Clifton. 2021. "Machine learning in patient flow: A review." *Prog. Biomed. Eng.* 3 (2): 022002. https://doi.org/10.1088/2516-1091/abddc5.

Fang, J., H. Hou, Z. M. Bi, D. Jin, L. Han, J. Yang, and S. Dai. 2021. "Data fusion in forecasting medical demands based on spectrum of post-earthquake diseases." *J. Ind. Inf. Integr.* 24 (Dec): 100235. https://doi.org/10.1016/j.jii.2021.100235.

Fei, L., and Y. Wang. 2022. "Demand prediction of emergency materials using case-based reasoning extended by the Dempster-Shafer theory." *Socio-Econ. Plann. Sci.* 84 (Dec): 101386. https://doi.org/10.1016/j.seps.2022.101386.

Gao, J., O. Murao, X. Pei, and Y. Dong. 2022. "Identifying evacuation needs and resources based on volunteered geographic information: A case of the rainstorm in July 2021, Zhengzhou, China." *Int. J. Environ. Res. Public Health* 19 (23): 16051. https://doi.org/10.3390/ijerph192316051.

Geng, S., H. Hou, and Z. Zhou. 2024. "A dynamic multi-objective model for emergency shelter relief system design integrating the supply and demand sides." *Nat. Hazard.* 120 (3): 2379–2402. https://doi.org/10.1007/s11069-023-06280-8.

Gilissen, H. K., M. Alexander, P. Matczak, M. Pettersson, and S. Bruzzone. 2016. "A framework for evaluating the effectiveness of flood emergency management systems in Europe." *Ecol. Soc.* 21 (4): 27. https://doi.org/10.5751/ES-08723-210427.

Guo, J., Q. Du, and Z. He. 2021. "A method to improve the resilience of multimodal transport network: Location selection strategy of emergency rescue facilities." *Comput. Ind. Eng.* 161 (Nov): 107678. https://doi.org/10.1016/j.cie.2021.107678.

Hamdi, A. H. A., E. Hatah, M. M. Bakry, A. H. Basari, and N. A. Hamdi. 2024. "Operations management of pharmaceutical supply during preparedness and disaster response: A scoping review." *Int. J. Disaster Risk Reduct.* 103 (Mar): 104296. https://doi.org/10.1016/j.ijdrr.2024.104296.

Hao, Y., M. Tian, Y. Wang, and M. Huang. 2023. "Demand forecasting for rush repair spare parts of power equipment using fuzzy C-means clustering and the fuzzy decision tree." *Int. J. Innovative Comput. Inf. Control* 19 (4): 1007–1021. https://doi.org/10.24507/ijicic.19.04.1007.

He, L. Y., H. Li, J. W. Bi, J. J. Yang, and Q. Zhou. 2022. "The impact of public health emergencies on hotel demand—Estimation from a new foresight perspective on the COVID-19." *Ann. Tourism Res.* 94 (May): 103402. https://doi.org/10.1016/j.annals.2022.103402.

Heydari, M., K. K. Lai, Y. Fan,, and X. Li. 2022. "A review of emergency and disaster management in the process of healthcare operation management for improving hospital surgical intake capacity." *Mathematics* 10 (15): 2784. https://doi.org/10.3390/math10152784.

Hezam, I. M. 2022. "COVID-19 global humanitarian response plan: An optimal distribution model for high-priority countries." *ISA Trans.* 124 (May): 1–20. https://doi.org/10.1016/j.isatra.2021.04.006.

Hu, L., W. Cong, M. Chen, J. Qiu, T. Yang, and Z. Wei. 2021a. "Prediction of power grid emergency material resources under meteorological disasters." In *Proc., 2021 IEEE IAS Industrial and Commercial Power System Asia (I and CPS Asia 2021)*, 1154–1159. New York: IEEE. https://doi.org/10.1109/ICPSAsia52756.2021.9621629.

Hu, M., Q. Liu, and H. Ai. 2021b. "Research on emergency material demand prediction model based on improved case-based reasoning and neural network." In *Proc., 2021 IEEE 3rd Int. Conf. on Civil Aviation Safety and Information Technology (ICCASIT)*, 270–276. New York: IEEE. https://doi.org/10.1109/ICCASIT53235.2021.9633410.

Huang, D., S. Wang, and Z. Liu. 2021. "A systematic review of prediction methods for emergency management." *Int. J. Disaster Risk Reduct.* 62 (Aug): 102412. https://doi.org/10.1016/j.ijdrr.2021.102412.

Huang, L., P. Shi, and H. Zhu. 2022. "An integrated urgency evaluation approach of relief demands for disasters based on social media data." *Int. J. Disaster Risk Reduct.* 80 (Oct): 103208. https://doi.org/10.1016/j.ijdrr.2022.103208.

Iqbal, U., P. Perez, and J. Barthelemy. 2021. "A process-driven and need-oriented framework for review of technological contributions to disaster management." *Heliyon* 7 (11): e08405. https://doi.org/10.1016/j.heliyon.2021.e08405.

Jebbor, S., C. Raddouane, and A. El Afia. 2022. "A preliminary study for selecting the appropriate AI-based forecasting model for hospital assets demand under disasters." *J. Humanitarian Logist. Supply Chain Manage.* 12 (1): 1–29. https://doi.org/10.1108/JHLSCM-12-2020-0123.

Jiang, J., J. Ma, and X. Chen. 2024. "Multi-regional collaborative mechanisms in emergency resource reserve and pre-dispatch design." *Int. J. Prod. Econ.* 270 (Apr): 109161. https://doi.org/10.1016/j.ijpe.2024.109161.

Jiang, Z., R. Ji, Y. Chen, and W. Ji. 2021. "Machine learning and simulation-based framework for disaster preparedness prediction." In *Proc., 2021 Winter Simulation Conf. (WSC)*, 1–10. New York: IEEE.

Jin, R., T. Xia, X. Liu, T. Murata, and K. S. Kim. 2021. "Predicting emergency medical service demand with bipartite graph convolutional networks." *IEEE Access* 9 (Jan): 9903–9915. https://doi.org/10.1109/ACCESS.2021.3050607.

Kamar, A., N. Maalouf, E. Hitti, G. E. Eid, H. Isma'eel, and I. H. Elhajj. 2021. "Challenge of forecasting demand of medical resources and supplies during a pandemic: A comparative evaluation of three surge calculators for COVID-19." *Epidemiol. Infect.* 149 (Jan): e51. https://doi.org/10.1017/S095026882100025X.

Kholaif, M. M. N. H. K., W. Gu, X. Tang, and W. H. Salama. 2024. "Post-emergencies opportunities for green supply chain management and urban supply chains' robustness; the moderating influence of blockchains as a sustainable urban resources management technique." *J. Cleaner Prod.* 475 (Oct): 143646. https://doi.org/10.1016/j.jclepro.2024.143646.

Kyrkou, C., P. Kolios, T. Theocharides, and M. Polycarpou. 2022. "Machine learning for emergency management: A survey and future outlook." *Proc. IEEE* 111 (1): 19–41. https://doi.org/10.1109/JPROC.2022.3223186.

La, R., L. Han, P. Bai, and Z. Zhang. 2022a. "Demand forecast of geological disaster rescue equipment based on 'scenario-task'." *Geotech. Geol. Eng.* 40 (4): 2267–2290. https://doi.org/10.1007/s10706-021-02025-1.

La, R., T. Lv, P. Bai, and Z. Zhang. 2022b. "Research on collaborative and optimal deployment and decision making among major geological disaster rescue subjects." *Geotech. Geol. Eng.* 40 (1): 57–71. https://doi.org/10.1007/s10706-021-01883-z.

Li, B., and Q. Liu. 2023. "Optimal scheduling of emergency materials based on gray prediction model under uncertain demand." *Electronics* 12 (20): 4337. https://doi.org/10.3390/electronics12204337.

Li, J. N., L. Shi, A. Q. Huang, Z. F. He, Y. X. Kang, and D. Li. 2023. "Forecasting emergency medicine reserve demand with a novel decomposition-ensemble methodology." *Complex Intell. Syst.* 9 (3): 2285–2295. https://doi.org/10.1007/s40747-021-00289-x.

Li, W., M. Zhang, B. Li, S. Li, and Z. Qiu. 2022a. "Rolling prediction of emergency supplies based on postdisaster multisource time-varying information." *Complexity* 2022 (1): 2431611. https://doi.org/10.1155/2022/2431611.

Li, Y., J. M. Hockenberry, J. Chen, and J. P. Cimiotti. 2022b. "Registered nurses: Can our supply meet the demand during a disaster?" *BMC Nurs.* 21 (1): 7. https://doi.org/10.1186/s12912-021-00794-x.

Liang, J., H. Zhao, Z. Yan, X. Mei, Y. Xue, and Y. Zhang. 2024. "Rapid construction method of emergency material supply chain based on shared platform covering market resources." *Int. J. Disaster Risk Reduct.* 105 (Apr): 104365. https://doi.org/10.1016/j.ijdrr.2024.104365.

Lin, A., H. Wu, G. Liang, A. Cardenas-Tristan, X. Wu, C. Zhao, and D. Li. 2020. "A big data-driven dynamic estimation model of relief supplies demand in urban flood disaster." *Int. J. Disaster Risk Reduct.* 49 (Oct): 101682. https://doi.org/10.1016/j.ijdrr.2020.101682.

Long, Y., G. Xu, J. Zhao, B. Xie, and M. Fang. 2023. "Dynamic truck–UAV collaboration and integrated route planning for resilient urban emergency response." *IEEE Trans. Eng. Manage.* 71 (Aug): 9826–9838. https://doi.org/10.1109/TEM.2023.3299693.

Ma, Q., Y. Zhou, and L. Liu. 2022. "Review and comparison of the demand analysis methods of maritime emergency resources." *Brodogradnja: Int. J. Nav. Archit. Ocean Eng. Res. Dev.* 73 (1): 141–162. https://doi.org/10.21278/brod73108.

Ma, Z., J. Zhang, and S. Gao. 2023. "Research on emergency material demand based on urgency and satisfaction under public health emergencies." *PLoS One* 18 (3): e0282796. https://doi.org/10.1371/journal.pone.0282796.

Mallouhy, R. E., C. Guyeux, C. A. Jaoude, and A. Makhoul. 2022. "Predicting fire brigades' operations based on their type of interventions." In *Proc., 2022 Int. Wireless Communications and Mobile Computing (IWCMC)*, 606–610. New York: IEEE. https://doi.org/10.1109/IWCMC55113.2022.9825380.

Martínez-Rojas, M., M. D. C. Pardo-Ferreira, and J. C. Rubio-Romero. 2018. "Twitter as a tool for the management and analysis of emergency situations: A systematic literature review." *Int. J. Inf. Manage.* 43 (Dec): 196–208. https://doi.org/10.1016/j.ijinfomgt.2018.07.008.

Marzook, M., M. Rajapaksha, T. Chanakya, and U. Dampage. 2023. "A mobile application that employs artificial intelligence to forecast the required number of beds for intensive care units in Sri Lanka." In *Proc., 2023 IEEE 17th Int. Conf. on Industrial and Information Systems (ICIIS)*, 221–226. New York: IEEE. https://doi.org/10.1109/ICIIS58898.2023.10253548.

Merz, B., et al. 2020. "Impact forecasting to support emergency management of natural hazards." *Rev. Geophys.* 58 (4): e2020RG000704. https://doi.org/10.1029/2020RG000704.

Nguyen, A., R. Pellerin, S. Lamouri, and B. Lekens. 2023. "Managing demand volatility of pharmaceutical products in times of disruption through news sentiment analysis." *Int. J. Prod. Res.* 61 (9): 2829–2840. https://doi.org/10.1080/00207543.2022.2070044.

Nguyen, L., Z. Yang, J. Li, Z. Pan, G. Cao, and F. Jin. 2022. "Forecasting people's needs in hurricane events from social network." *IEEE Trans. Big Data* 8 (1): 229–240. https://doi.org/10.1109/TBDATA.2019.2941887.

Oktarina, R., S. N. Bahagia, L. Diawati, and K. S. Pribadi. 2020. "Artificial neural network for predicting earthquake casualties and damages in Indonesia." *IOP Conf. Ser.: Earth Environ. Sci.* 426 (1): 012156. https://doi.org/10.1088/1755-1315/426/1/012156.

Ramgopal, S., T. Westling, N. Siripong, D. D. Salcido, and C. Martin-Gill. 2021. "Use of a metalearner to predict emergency medical services demand in an urban setting." *Comput. Methods Programs Biomed.* 207 (Aug): 106201. https://doi.org/10.1016/j.cmpb.2021.106201.

Shao, J., C. Liang, X. Wang, X. Wang, and L. Liang. 2020. "Relief demand calculation in humanitarian logistics using material classification." *Int. J. Environ. Res. Public Health* 17 (2): 582. https://doi.org/10.3390/ijerph17020582.

Song, Q., Y. J. Zheng, J. Yang, Y. J. Huang, W. G. Sheng, and S. Y. Chen. 2023. "Predicting demands of COVID-19 prevention and control materials via co-evolutionary transfer learning." *IEEE Trans. Cybern.* 53 (6): 3859–3872. https://doi.org/10.1109/TCYB.2022.3164412.

Tabata, T., A. Onishi, T. Saeki, and P. Tsai. 2019. "Earthquake disaster waste management reviews: Prediction, treatment, recycling, and prevention." *Int. J. Disaster Risk Reduct.* 36 (May): 101119. https://doi.org/10.1016/j.ijdrr.2019.101119.

Tan, L., J. Guo, S. Mohanarajah, and K. Zhou. 2021. "Can we detect trends in natural disaster management with artificial intelligence? A review of modeling practices." *Nat. Hazard.* 107 (3): 2389–2417. https://doi.org/10.1007/s11069-020-04429-3.

Tian, S., and Y. Mei. 2023. "Emergency regional food supply chain design and its labor demand forecasting model: Application to COVID-19 pandemic disruption." *Front. Sustainable Food Syst.* 7 (Sep): 1189451. https://doi.org/10.3389/fsufs.2023.1189451.

Toland, J. C., A. M. Wein, A. M. Wu, and L. A. Spearing. 2023. "A conceptual framework for estimation of initial emergency food and water resource requirements in disasters." *Int. J. Disaster Risk Reduct.* 90 (May): 103661. https://doi.org/10.1016/j.ijdrr.2023.103661.

Vollmer, M. A. C., B. Glampson, T. Mellan, S. Mishra, L. Mercuri, C. Costello, R. Klaber, G. Cooke, S. Flaxman, and S. Bhatt. 2021. "A unified machine learning approach to time series forecasting applied to demand at emergency departments." *BMC Emerg. Med.* 21 (1): 9. https://doi.org/10.1186/s12873-020-00395-y.

Wang, S., Y. J. Wu, and R. Li. 2022. "An improved genetic algorithm for location allocation problem with grey theory in public health emergencies." *Int. J. Environ. Res. Public Health* 19 (15): 9752. https://doi.org/10.3390/ijerph19159752.

Wang, Z., and H. Jia. 2023. "An intuitionistic fuzzy grey-Markov method with application to demand forecasting for emergency supplies during major epidemics." *Grey Syst. Theory Appl.* 14 (1): 185–208. https://doi.org/10.1108/GS-07-2023-0062.

Yan, K., Y. Jiang, J. Qiu, X. Zhong, Y. Wang, J. Deng, J. Lian, T. Wang, and C. Cao. 2017. "The equity of China's emergency medical services from 2010–2014." *Int. J. Equity Health* 16 (1): 10. https://doi.org/10.1186/s12939-016-0507-5.

Yang, M., Y. Liu, and G. Yang. 2021. "Multi-period dynamic distributionally robust pre-positioning of emergency supplies under demand uncertainty." *Appl. Math. Modell.* 89 (Jan): 1433–1458. https://doi.org/10.1016/j.apm.2020.08.035.

Zhang, T., S. Shen, C. Cheng, K. Su, and X. Zhang. 2021. "A topic model based framework for identifying the distribution of demand for relief supplies using social media data." *Int. J. Geogr. Inf. Sci.* 35 (11): 2216–2237. https://doi.org/10.1080/13658816.2020.1869746.

Zhang, X., J. Yu, Y. Chen, J. Wen, J. Chen, and Z. Yin. 2020. "Supply–demand analysis of urban emergency shelters based on spatiotemporal population estimation." *Int. J. Disaster Risk Sci.* 11 (4): 519–537. https://doi.org/10.1007/s13753-020-00284-9.

Zheng, Y. J., S. L. Yu, Q. Song, Y. J. Huang, W. G. Sheng, and S. Y. Chen. 2022. "Co-evolutionary fuzzy deep transfer learning for disaster relief demand forecasting." *IEEE Trans. Emerging Top. Comput.* 10 (3): 1361–1373. https://doi.org/10.1109/TETC.2021.3085337.

Zhu, X., D. Regan, and B. Sun. 2022. "Does exploring the characteristics of emergency supplies really matter for disaster response operations?" *Nat. Hazard.* 110 (1): 175–189. https://doi.org/10.1007/s11069-021-04941-0.

Zhu, X., G. Zhang, and B. Sun. 2019. "A comprehensive literature review of the demand forecasting methods of emergency resources from the perspective of artificial intelligence." *Nat. Hazard.* 97 (1): 65–82. https://doi.org/10.1007/s11069-019-03626-z.